



A unified computational framework for developing general multi-time-step (MTS) coupling schemes for time-dependent problems

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Abstract

Multi-time-step (MTS) methods solve temporal multi-scale problems by decomposing the spatial problem domain into multiple subdomains and integrating each subdomain with a time step that is suitable for capturing its response. A critical limitation of existing MTS methods, however, is that they are based on enforcing continuity of velocities across the interface between subdomains and consequently lead to erroneous drifts in displacements that show up as nonphysical gaps and overlaps in the solution. In this article, we propose a new MTS framework that enables simultaneous enforcement of multiple continuity constraints between subdomains and the use of general interpolation schemes for associated Lagrange multipliers. To make the framework as general as possible, we first express almost all existing time integration schemes in a Generalized Runge–Kutta–Nyström (GRKN) form, which then serve as the underlying time integration schemes for individual subdomains. MTS methods designed using the proposed framework are able to achieve unconditional stability by introducing a scalar auxiliary variable (SAV) based on physical energy of the system and ensuring that this energy remains bounded during the simulation. We design new MTS methods based on piecewise-linear and quadratic interpolations of Lagrange multipliers with simultaneous enforcement of velocity and displacement continuity (VCDC) to demonstrate typical use-cases of the proposed framework. We present several numerical examples to show that solutions obtained using the new VDCDC MTS methods are able to achieve greater accuracy for the same computational cost in comparison to single time integration schemes and existing MTS methods.

Keywords Transient structural mechanics · Coupling methods · Multiple time steps · Time integration · Multi-scale · Energy stability

1 Introduction

Problems in structural dynamics and wave propagation are usually governed by hyperbolic partial differential equations. Due to the lack of general analytical solutions to such problems, they are usually solved using techniques such as modal decomposition, space-time methods, and time integration of semi-discrete systems [1–4]. In this paper, we focus on time integration methods where the problem domain is first discretized in space, leading to a

semi-discrete system of ordinary differential equations, and using a time integration scheme to advance the solution in time. There are numerous time integration schemes in the literature with widely different characteristics of stability, accuracy, and computational cost [5–30]. Broadly, time integration schemes may be categorized into linear multi-step (LMS) and Runge–Kutta (RK) methods. LMS schemes employ known state vectors from a few previous time steps to advance the solution to the next time step. One of the most well-known LMS schemes is the Newmark scheme [5] that uses two parameters γ and β and leads to several implicit and explicit time integration methods. To improve the performance of Newmark schemes with respect to spurious high-frequency oscillations, the CH- α [10] scheme was proposed. Although CH- α schemes introduce numerical damping while maintaining second-order accuracy for the displacement and velocity, they are usually first-order accurate for the acceleration [31] and exhibit overshooting

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for large time steps [32]. Recently, Zhou and Tamma [14] proposed generalized single-step single-solver (GS4) approach and showed their spectral equivalence with existing LMS schemes. On the other hand, RK schemes divide each time step into multiple sub-stages and compute intermediate state vectors at these stages to advance the solution [33–35]. Note that composite time integration schemes that divide a single time step into two or more sub-steps and use different schemes for each of the sub-steps to advance the solution also belong to the class of RK methods [36]. In particular, Bathe and his co-workers proposed composite time integration schemes [16–21] that use two sub-steps to achieve greater accuracy than LMS schemes. Similarly, there are several other composite time integration schemes that reduce dispersion and dissipation errors [22–30].

Despite the advances in time integration methods, problems that span multiple temporal scales still pose a big challenge because no single time integration method that uses a uniform time step (UTS) over the entire problem domain is able to capture the solution with sufficient accuracy without incurring a disproportionate computational cost penalty. For example, simulating the response of a structure subject to a local impact would require a fine spatial and temporal resolution in the vicinity of the impact whereas regions away from the impact zone may be solved with a coarse discretization. Using the UTS approach for such problems, one would have to use a fine mesh and a proportionately small time step to capture the fine-scale response which would invariably lead to a large computational cost. Instead, using a multi-time-step (MTS) approach that allows one to decompose the problem domain into multiple subdomains and enables the use of different time integration schemes and different time steps in different subdomains, is better suited for such problems. By tailoring the spatial and temporal discretization to the mechanical characteristics of each individual subdomain, MTS methods are able to achieve high overall accuracy at low computational cost. There are several MTS methods in the literature for LMS schemes [37–56]. Among these, methods that use dual-Schur domain decomposition were initially proposed by Gravouil and Combescure [44, 45] for coupling Newmark schemes, which was subsequently improved by Prakash and his coworkers [46, 47] to eliminate numerical dissipation (referred to as the PH method here). In [53–55], a heterogeneous asynchronous time integration approach was developed to couple the Simo, Krenk, Verlet, HHT- α , WBZ- α , and CH- α schemes as well as Newmark schemes. More recently, Tae and Tamma [56] presented a differential algebraic equation (DAE) framework for coupling different GS4 time integration schemes using multiple time steps (referred to as the DAE-GS4 method here).

Unfortunately, all existing MTS methods suffer from a critical drawback which is that they all enforce continuity of a single kinematic quantity (most commonly velocity

continuity) across the interface between subdomains. This invariably leads to drifts in displacement that manifest as nonphysical gaps and overlaps between the subdomains. The inability to simultaneously impose continuity of displacements, velocities, and accelerations between the subdomains can lead to large errors in the solution, as shown in Sect. 4.2. Another key idea with almost all existing MTS methods is that kinematic quantities and/or the Lagrange multipliers used to enforce continuity constraints are assumed to vary *linearly* over the largest time step among the subdomains. While these choices lead to stable coupling methods, they severely restrict the applicability of the MTS approach.

In this paper, we present a new MTS framework for existing time integration schemes that allows simultaneous enforcement of multiple continuity constraints to ensure zero drift in displacements and velocities, while maintaining unconditional stability, high accuracy, and low computational cost. To incorporate as many time integration schemes as possible, we extend the Runge–Kutta–Nyström (RKN) formulation by expressing it in a more general form that not only encompasses all existing RKN schemes but also all GS4 schemes that have been shown to be spectrally equivalent to LMS schemes. We call these time integration schemes the Generalized Runge–Kutta–Nyström (GRKN) schemes and in Sect. 2 and Appendix A, we show how the GRKN schemes encompass most existing time integration schemes. In Sect. 3, we formulate a general MTS framework to enable coupling two or more GRKN schemes with multiple time steps. In addition to enforcing multiple continuity constraints to ensure zero drift, the framework introduces a scalar auxiliary variable (SAV) to ensure unconditional stability. We discuss a computational cost model for the resulting coupled algorithm that may be used to estimate potential computational gains of using the MTS approach. Finally, in Sect. 4, we demonstrate several use-cases of the proposed MTS framework and evaluate its performance with numerical examples.

2 Time integration schemes: generalized Runge–Kutta–Nyström scheme

Runge–Kutta–Nyström (RKN) schemes [35] have been used for second-order ordinary differential equations (ODEs) unlike RK schemes [57–59], which have been developed to solve first-order ODEs. Note that the RKN schemes encompass all RK schemes for second-order ODEs and almost all composite time integration schemes as well [36]. However, as shown in Fig. 1, RKN schemes do not encompass all GS4 schemes. For instance, Wilson- θ , HHT- α , WBZ- α , and CH- α are examples of GS4 schemes that are not

Fig. 1 Classification of time integration schemes for second-order ODEs

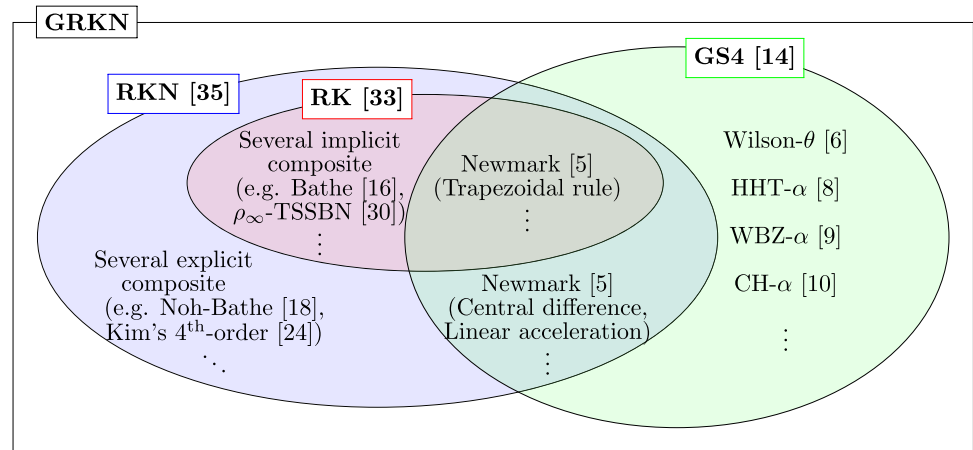
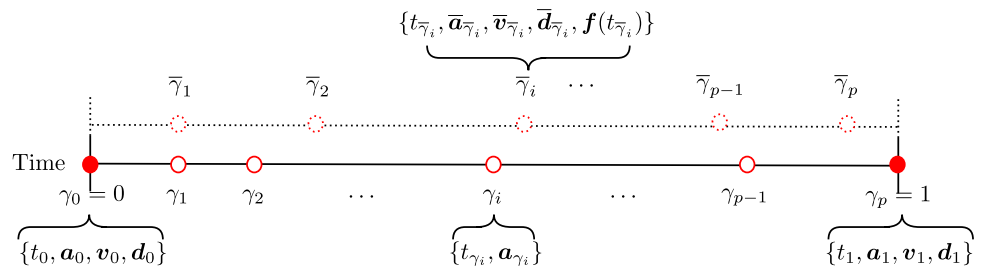


Fig. 2 Representation of p -stage GRKN scheme



encompassed by RKN schemes. In this section, we present a Generalized Runge–Kutta–Nyström (GRKN) scheme that is able to encompass all RKN and GS4 schemes.

The semi-discrete governing differential equations in time t for nonlinear structural dynamics along with their initial and boundary conditions are written as:

$$M\ddot{\mathbf{u}}(t) + \mathbf{p}(\dot{\mathbf{u}}(t), \mathbf{u}(t)) = \mathbf{f}(t) \quad \text{in } \Omega \quad \forall t \in (0, T] \quad (1)$$

$$\mathbf{u}(0) = \mathbf{d}_0, \quad \dot{\mathbf{u}}(0) = \mathbf{v}_0 \quad \text{in } \Omega \quad \text{at } t = 0 \quad (2)$$

$$\mathbf{u}(t) = \mathbf{d}_D(t), \quad \dot{\mathbf{u}}(t) = \mathbf{v}_D(t) \quad \text{on } \Gamma_D \quad \forall t \in (0, T] \quad (3)$$

where M is the mass matrix, $\mathbf{u}$, $\mathbf{p}$ and $\mathbf{f}$ denote the displacement, internal force and external force vectors, respectively, and T is the time duration. An overdot denotes time derivative, $\mathbf{d}_0$ and $\mathbf{v}_0$ represent initial displacement and velocity, respectively, in the problem domain Ω and $\mathbf{d}_D$ and $\mathbf{v}_D$ denote prescribed displacement and velocity on the Dirichlet boundary Γ_D .

For time discretization, a p -stage GRKN scheme is shown in Fig. 2 to advance state vectors from t_0 to $t_1 = t_0 + \Delta t$ (where Δt is the time step) through sub-stages $i = 1, 2, \dots, p$. Accelerations, velocities, displacements, and external forces at these sub-stages are approximated as $\mathbf{a}$, $\mathbf{v}$, $\mathbf{d}$, and $\mathbf{f}$, respectively and the subscript denotes the time instant with which each quantity is associated. Note that there are two sets of time instants at each sub-stage: (1) $t_{\bar{\gamma}_i} = t_0 + \bar{\gamma}_i \Delta t$ where intermediate algorithmic quantities

$\bar{(\cdot)}$ are computed; and (2) $t_{\gamma_i} = t_0 + \gamma_i \Delta t$ where sub-stage accelerations $\mathbf{a}_{\gamma_i}$ are computed. Constants $\bar{\gamma}_i$ and γ_i are algorithmic parameters.

Upon time discretization and linearization, the semi-discrete system of equations (1) is transformed into a system of algebraic equations that are solved using a standard iterative procedure [3, 4, 60]. For ease of exposition, we only show the time integration algorithm for a typical iteration here. At the i^{th} sub-stage, algorithmic quantities $\bar{\mathbf{a}}_{\bar{\gamma}_i}$, $\bar{\mathbf{v}}_{\bar{\gamma}_i}$, and $\bar{\mathbf{d}}_{\bar{\gamma}_i}$ are computed using the following time integrator:

$$M\bar{\mathbf{a}}_{\bar{\gamma}_i} + D_{\bar{\gamma}_i}\bar{\mathbf{v}}_{\bar{\gamma}_i} + K_{\bar{\gamma}_i}\bar{\mathbf{d}}_{\bar{\gamma}_i} = \mathbf{f}(t_{\bar{\gamma}_i}) \quad \text{for } i = 1, 2, \dots, p \quad (4)$$

$$\bar{\mathbf{a}}_{\bar{\gamma}_i} = \sum_{j=0}^p \bar{\alpha}_{ij} \mathbf{a}_{\gamma_j} \quad (5)$$

$$\bar{\mathbf{v}}_{\bar{\gamma}_i} = \mathbf{v}_0 + \Delta t \sum_{j=0}^p \bar{\nu}_{ij} \mathbf{a}_{\gamma_j} \quad (6)$$

$$\bar{\mathbf{d}}_{\bar{\gamma}_i} = \mathbf{d}_0 + \bar{\delta}_i \Delta t \mathbf{v}_0 + \Delta t^2 \sum_{j=0}^p \bar{\mu}_{ij} \mathbf{a}_{\gamma_j} \quad (7)$$

where $D_{\bar{\gamma}_i}$ and $K_{\bar{\gamma}_i}$ are damping and stiffness matrices, respectively, linearized about time $t_{\bar{\gamma}_i}$. Constants $\bar{\alpha}_{ij}$, $\bar{\nu}_{ij}$, $\bar{\delta}_i$, $\bar{\mu}_{ij}$, denote algorithmic parameters of time integration. Note that i varies between $1, 2, \dots, p$ and j varies between $0, 1, \dots, p$ resulting in $3p(p+1) + p = 3p^2 + 4p$

parameters. With an additional p parameters due to $\bar{\gamma}_i$, we get $3p^2 + 5p$ algorithmic parameters. Note that γ_i are not treated as independent parameters as they can be expressed in terms of $\bar{\gamma}_i$ and $\bar{\alpha}_{ij}$.

Equations (5)–(7) may be substituted into Eq. (4) to compute intermediate acceleration vectors $\mathbf{a}_{\gamma_i}$ and update final sub-stage velocity and displacement as:

$$\sum_{j=0}^p M_{ij} \mathbf{a}_{\gamma_j} = \mathbf{f}(t_{\bar{\gamma}_i}) - (N_i \mathbf{v}_0 + \mathbf{K}_{\bar{\gamma}_i} \mathbf{d}_0) \quad \text{for } i = 1, 2, \dots, p \quad (8)$$

$$\mathbf{v}_1 = \mathbf{v}_0 + \Delta t \sum_{j=0}^p \nu_j \mathbf{a}_{\gamma_j} \quad (9)$$

$$\mathbf{d}_1 = \mathbf{d}_0 + \delta_p \Delta t \mathbf{v}_0 + \Delta t^2 \sum_{j=0}^p \mu_j \mathbf{a}_{\gamma_j} \quad (10)$$

where matrices $M_{ij} = \bar{\alpha}_{ij} \mathbf{M} + \bar{\nu}_{ij} \Delta t \mathbf{D}_{\bar{\gamma}_i} + \bar{\mu}_{ij} \Delta t^2 \mathbf{K}_{\bar{\gamma}_i}$ and $N_i = \mathbf{D}_{\bar{\gamma}_i} + \bar{\delta}_i \Delta t \mathbf{K}_{\bar{\gamma}_i}$. Note that additional algorithmic parameters ν_j , δ_p , and μ_j are introduced above taking the total number of algorithmic parameters to $3p^2 + 7p + 3$.

Equation (8) may be rewritten compactly as:

$$\mathbb{M} \mathbf{a}_1 = \mathbb{f}_{0 \rightarrow 1} - \mathbb{N} \mathbf{x}_0 \quad (11)$$

where block matrices and vectors $\mathbb{M}$, $\mathbb{N}$, $\mathbf{a}_1$, $\mathbb{f}_{0 \rightarrow 1}$, and $\mathbf{x}_0$ along with the sizes are defined as:

$$\begin{aligned} \mathbb{M} &= \begin{bmatrix} M_{11} & M_{12} & \cdots & M_{1p} \\ M_{21} & M_{22} & \cdots & M_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ M_{p1} & M_{p2} & \cdots & M_{pp} \end{bmatrix}_{p \times p}, \\ \mathbf{a}_1 &= \begin{bmatrix} \mathbf{a}_{\gamma_1} \\ \mathbf{a}_{\gamma_2} \\ \vdots \\ \mathbf{a}_1 \end{bmatrix}_{p \times 1}, \quad \mathbb{f}_{0 \rightarrow 1} = \begin{bmatrix} \mathbf{f}(t_{\bar{\gamma}_1}) \\ \mathbf{f}(t_{\bar{\gamma}_2}) \\ \vdots \\ \mathbf{f}(t_{\bar{\gamma}_p}) \end{bmatrix}_{p \times 1}, \\ \mathbb{N} &= \begin{bmatrix} M_{10} & N_1 & K_{\bar{\gamma}_1} \\ M_{20} & N_2 & K_{\bar{\gamma}_2} \\ \vdots & \vdots & \vdots \\ M_{p0} & N_p & K_{\bar{\gamma}_p} \end{bmatrix}_{p \times 3}, \\ \mathbf{x}_0 &= \begin{bmatrix} \mathbf{a}_0 \\ \mathbf{v}_0 \\ \mathbf{d}_0 \end{bmatrix}_{3 \times 1} \end{aligned} \quad (12)$$

Further, Eqs. (9) and (10) may be combined with Eq. (11) to obtain:

$$\mathbf{x}_1 = \mathbb{D} \mathbf{x}_0 + \mathbb{P} \mathbf{a}_1 \quad (13)$$

where block matrices and vectors $\mathbf{x}_1$, $\mathbb{D}$, and $\mathbb{P}$ are defined as:

$$\begin{aligned} \mathbf{x}_1 &= \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{v}_1 \\ \mathbf{d}_1 \end{bmatrix}_{3 \times 1}, \quad \mathbb{D} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \nu_0 \Delta t \mathbf{I} & \mathbf{I} & \mathbf{0} \\ \mu_0 \Delta t^2 \mathbf{I} & \delta_p \mathbf{I} & \mathbf{I} \end{bmatrix}_{3 \times 3}, \\ \mathbb{P} &= \begin{bmatrix} \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{I} \\ \nu_1 \Delta t \mathbf{I} & \nu_2 \Delta t \mathbf{I} & \cdots & \nu_{p-1} \Delta t \mathbf{I} & \nu_p \Delta t \mathbf{I} \\ \mu_1 \Delta t^2 \mathbf{I} & \mu_2 \Delta t^2 \mathbf{I} & \cdots & \mu_{p-1} \Delta t^2 \mathbf{I} & \mu_p \Delta t^2 \mathbf{I} \end{bmatrix}_{3 \times p} \end{aligned} \quad (14)$$

Note that $\mathbf{I}$ denotes the identity matrix of the size of the number of degrees of freedom in the problem. Finally, substituting Eq. (11) into Eq. (13), we obtain the conventional time-stepping form of GRKN schemes as:

$$\mathbf{x}_1 = \mathbb{G} \mathbf{x}_0 + \mathbb{L} \mathbb{f}_{0 \rightarrow 1} \quad (15)$$

where $\mathbb{G} = \mathbb{D} - \mathbb{L} \mathbb{N}$ and $\mathbb{L} = \mathbb{P} \mathbb{M}^{-1}$.

To show equivalence of the GRKN schemes to existing time integration schemes, its parameters may be conveniently listed in a Butcher tableau shown in Table 1. Recall that GRKN schemes comprise $3p^2 + 7p + 3$ parameters. For instance, when $p = 1$ (i.e. single-stage schemes), the number of parameters is 13. When $p = 2$ (most composite schemes), the number of parameters is 29, and so on. Note, however, that for common time integration schemes, most of these parameters are 0 (see Appendix A).

Note that an s -stage RKN scheme [35] with parameters a_{ij} , a_{ij}^* , b_i , b_i^* , and c_i can be expressed as a p -stage GRKN scheme (with $p = s + 1$) as shown in Table 2. Since RKN schemes include most existing explicit and implicit composite time integration schemes for structural dynamics (see [36] for details), GRKN schemes also encompass them. Furthermore, a RKN scheme with the following relationships reduces to an s -stage Runge-Kutta (RK) scheme [33]:

$$a_{ir}^* = \sum_{j=1}^s a_{ij} a_{jr}, \quad b_j^* = \sum_{i=1}^s b_i a_{ij} \quad \text{for } i = 1, 2, \dots, s \quad (16)$$

Note that RK schemes with $a_{sj} = b_j$ ($j = 1, 2, \dots, s$) are stiffly accurate [61] and can achieve BN-stability [62] if the following relationship is satisfied: 1) $b_i \geq 0$ for $i = 1, 2, \dots, s$;

Table 1 The Butcher tableau for a p -stage GRKN scheme

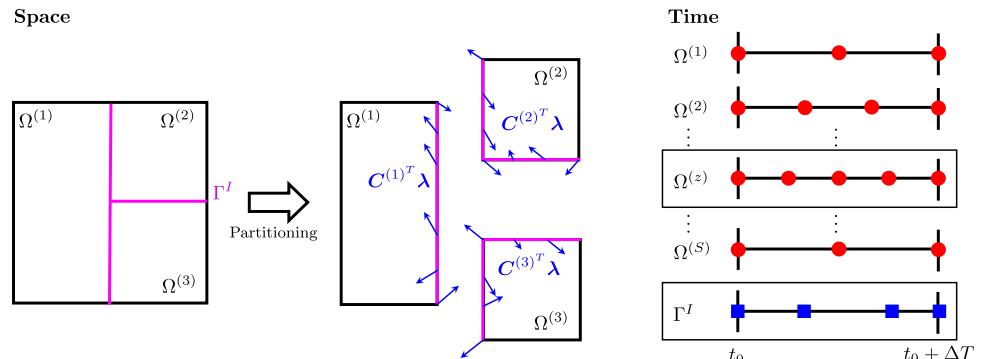
f	$\bar{a}$				$\bar{v}$				$\bar{d}$				
$\bar{\gamma}_1$	$\bar{\alpha}_{10}$	$\bar{\alpha}_{11}$	$\cdots$	$\bar{\alpha}_{1p}$	$\bar{\nu}_{10}$	$\bar{\nu}_{11}$	$\cdots$	$\bar{\nu}_{1p}$	$\bar{\delta}_1$	$\bar{\mu}_{10}$	$\bar{\mu}_{11}$	$\cdots$	$\bar{\mu}_{1p}$
$\bar{\gamma}_2$	$\bar{\alpha}_{20}$	$\bar{\alpha}_{21}$	$\cdots$	$\bar{\alpha}_{2p}$	$\bar{\nu}_{20}$	$\bar{\nu}_{21}$	$\cdots$	$\bar{\nu}_{2p}$	$\bar{\delta}_2$	$\bar{\mu}_{20}$	$\bar{\mu}_{21}$	$\cdots$	$\bar{\mu}_{2p}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
$\bar{\gamma}_p$	$\bar{\alpha}_{p0}$	$\bar{\alpha}_{p1}$	$\cdots$	$\bar{\alpha}_{pp}$	$\bar{\nu}_{p0}$	$\bar{\nu}_{p1}$	$\cdots$	$\bar{\nu}_{pp}$	$\bar{\delta}_p$	$\bar{\mu}_{p0}$	$\bar{\mu}_{p1}$	$\cdots$	$\bar{\mu}_{pp}$
					ν_0	ν_1	$\cdots$	ν_p	δ_p	μ_0	μ_1	$\cdots$	μ_p
					v				d				

Table 2 The Butcher tableau for an s -stage RKN scheme [35] expressed in GRKN form

c_1	0	1	0	$\cdots$	0	0	a_{11}	$\cdots$	a_{1s}	0	c_1	0	a_{11}^*	$\cdots$	a_{1s}^*	0	
c_2	0	0	1	$\cdots$	0	0	a_{21}	$\cdots$	a_{2s}	0	c_2	0	a_{21}^*	$\cdots$	a_{2s}^*	0	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
c_s	0	0	0	$\cdots$	1	0	a_{s1}	$\cdots$	a_{ss}	0	c_s	0	a_{s1}^*	$\cdots$	a_{ss}^*	0	
1	0	0	0	$\cdots$	0	1	b_1	$\cdots$	b_s	0	1	0	b_1^*	$\cdots$	b_s^*	0	
							0	b_1	$\cdots$	b_s	0	1	0	b_1^*	$\cdots$	b_s^*	0

Table 3 The Butcher tableau for the GS4 scheme [14] expressed in GRKN form

W_1	$1 - W_1\Lambda_6$	$W_1\Lambda_6$	$W_1\Lambda_4 - W_2\Lambda_5$	$W_2\Lambda_5$	$W_1\Lambda_1$	$W_2\Lambda_2 - W_3\Lambda_3$	$W_3\Lambda_3$
			$\lambda_4 - \lambda_5$	λ_5	λ_1	$\lambda_2 - \lambda_3$	λ_3

Fig. 3 Spatial domain decomposition of a problem into multiple subdomains that are integrated using different GRKN schemes and different time steps. A single block time step ΔT is used to advance the coupled system of subdomains

and 2) matrix $\mathbf{R}$ is non-negative definite where the components of $\mathbf{R}$ are defined as $R_{ij} = b_i a_{ij} + b_j a_{ji} - b_i b_j$.

Note that GRKN schemes with parameters shown in Table 3 lead to GS4 schemes [14], which encompass several common time integration schemes including Newmark [5], Wilson- θ [6], HHT- α [8], WBZ- α [9], CH- α [10], U0 ($\rho_\infty^{\min}, \rho_\infty^{\max}, \rho_\infty^s$) [63], and V0 ($\rho_\infty^{\min}, \rho_\infty^{\max}, \rho_\infty^s$) [64] schemes. Appendix A shows the GRKN Butcher tableau for several LMS and composite schemes.

3 Multi-time-step framework for coupling different GRKN time integration schemes

In this section, we present a novel, unconditionally stable MTS framework for GRKN schemes that allows one to decompose a problem domain into S subdomains $\Omega^{(z)}$ (where the superscript $z \in \{1, 2, \dots, S\}$ represents the subdomain index) with a common interface Γ^I and enables the use of different GRKN schemes and/or different time steps in each subdomain (see Fig. 3). The semi-discrete system of equations corresponding to such a decomposed problem domain is given by:

$$\begin{aligned} \mathbf{M}^{(z)} \ddot{\mathbf{u}}^{(z)}(t) + \mathbf{p}^{(z)}(\mathbf{u}^{(z)}(t), \dot{\mathbf{u}}^{(z)}(t)) + \mathbf{C}^{(z)T} \boldsymbol{\lambda}(t) &= \mathbf{f}^{(z)}(t) \\ \sum_{z=1}^S \mathbf{C}^{(z)} \mathbf{u}^{(z)}(t) &= \mathbf{0} \end{aligned} \quad (17)$$

where $\boldsymbol{\lambda}$ denotes the Lagrange multipliers, and $\mathbf{C}^{(z)}$ is a connectivity matrix between subdomain degrees of freedom and interface degrees of freedom. Initial and boundary conditions for the decomposed system may be obtained by projecting Eqs. (2) and (3) for the original problem domain onto its subdomains. For further details of the spatial discretization and interpolation of subdomains and the Lagrange multiplier fields, one may refer to [42–49]. Note that subdomain discretizations that do not match across the interface Γ^I may be coupled using mortar methods [65] and variationally consistent coupling (VCC) [66].

For temporal discretization, subdomain $\Omega^{(z)}$ is advanced with time step $\Delta t^{(z)}$ shown as red circles such that the system time step $\Delta T = m^{(z)} \Delta t^{(z)}$ where $m^{(z)}$ is the time-step ratio for subdomain $\Omega^{(z)}$. Coupling of subdomains across the interface is achieved by enforcing multiple simultaneous continuity constraints using Lagrange multipliers evaluated at discrete times (shown with solid blue squares in Fig. 3). Solution of this coupled system of subdomains is achieved using the proposed MTS framework that entails four key steps: (1) solution of the individual subdomains at their respective time steps; (2) computation of Lagrange multipliers over the interface; (3) update of subdomain solutions to enforce multiple simultaneous continuity constraints; and (4) stabilization using a scalar auxiliary variable (SAV). In the following subsections, these four steps are described for an arbitrary subdomain $\Omega^{(z)}$ and its coupling with the interface Γ^I . For simplicity, we assume the

3.1 Temporal discretization of subdomain and interface quantities

To facilitate the development of the proposed MTS framework, we first briefly describe the notation that we use for representing the subdomain and interface quantities at different instants of time within the system time step ΔT . The timelines for a typical subdomain $\Omega^{(z)}$ and the interface Γ^I are shown in Fig. 4. Assuming a time-step ratio of $m^{(z)}$, the solid red circles in Fig. 4 represent the time steps for subdomain $\Omega^{(z)}$ corresponding to times $t_\ell^{(z)}$ with $\ell = 0, 1, 2, \dots, m^{(z)}$. The subdomain state vectors $\mathbf{x}_\ell^{(z)}$ at these time steps are computed in a manner similar to the underlying GRKN scheme shown in Fig. 2 – by enforcing the equations of motion for subdomain $\Omega^{(z)}$ at the times represented by the dashed red circles i.e. $t_{\ell+\bar{\gamma}_i}^{(z)}$ where $\bar{\gamma}_i$ represent algorithmic parameters of the GRKN scheme. Note that there may be multiple GRKN sub-stages within each time step $\Delta t^{(z)}$, however in Fig. 4, we depict only a 1-stage scheme for simplicity.

Lagrange multipliers λ_β on the interface Γ^I depicted by solid blue squares at times t_β in Fig. 4 are computed by enforcing multiple continuity constraints at times $t_{\bar{\beta}}$ represented with blue dashed squares. The values of β and $\bar{\beta}$ may be chosen from the following sets: $\beta \in \mathcal{B}$ where $\mathcal{B} := \{\beta_1, \beta_2, \dots, \beta_{p^I}\}$ and $\bar{\beta} \in \bar{\mathcal{B}}$ where $\bar{\mathcal{B}} := \{\bar{\beta}_1, \bar{\beta}_2, \dots, \bar{\beta}_{p^I}\}$ depending upon the number of continuity constraints enforced, p^I , and the instants at which they are enforced, $t_{\bar{\beta}}$. Different choices for values in $\mathcal{B}$ and $\bar{\mathcal{B}}$ lead to different MTS methods and to different numerical properties, as discussed later in Sect. 4. However, one must ensure that $\mathcal{B}$ consists of distinct nonzero values so that the resulting system of equations is well-posed.

Since the discretization of subdomain state vectors $\mathbf{x}_\ell^{(z)}$ and the Lagrange multipliers λ_β may be non-matching, in general, we need to be able to transfer information between individual subdomains and the interface at any instant of time t . To allow for such exchange of information, the

subdomain solution $\mathbf{x}^{(z)}(t)$ at any time t may be obtained by interpolation as:

$$\mathbf{x}^{(z)}(t) \equiv \sum_{\ell=0}^{m^{(z)}} \mathbb{W}_\ell^{(z)}(t) \mathbf{x}_\ell^{(z)} \quad (18)$$

where $\mathbb{W}_\ell^{(z)}(t)$ is a matrix of user-defined interpolation functions $w_\ell^a(t)$, $w_\ell^v(t)$, and $w_\ell^d(t)$ that interpolate accelerations, velocities, and displacements of subdomain $\Omega^{(z)}$ as:

$$\begin{aligned} \mathbf{a}^{(z)}(t) &= \sum_{\ell=0}^{m^{(z)}} w_\ell^a(t) \mathbf{a}_\ell^{(z)}, \\ \mathbf{v}^{(z)}(t) &= \sum_{\ell=0}^{m^{(z)}} w_\ell^v(t) \mathbf{v}_\ell^{(z)}, \\ \mathbf{d}^{(z)}(t) &= \sum_{\ell=0}^{m^{(z)}} w_\ell^d(t) \mathbf{d}_\ell^{(z)} \end{aligned} \quad (19)$$

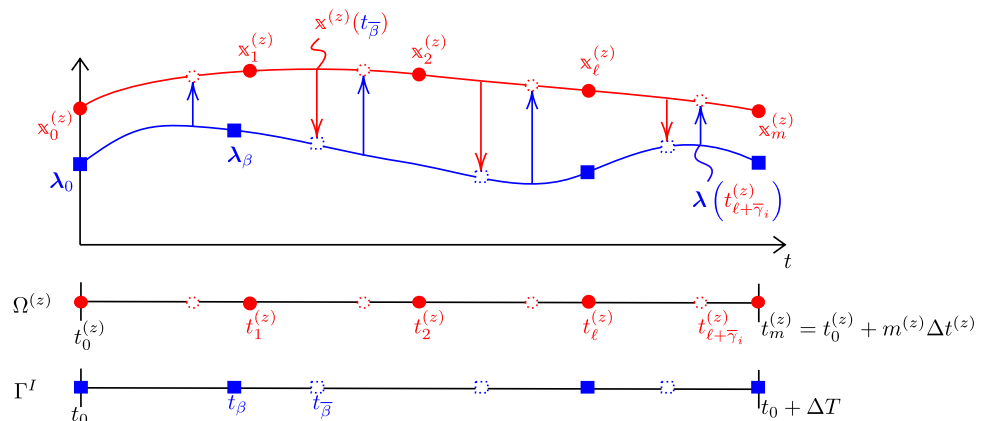
Note $\mathbb{W}_\ell^{(z)}(t) = \text{diag} \left(w_\ell^a(t) \mathbf{I}^{(z)}, w_\ell^v(t) \mathbf{I}^{(z)}, w_\ell^d(t) \mathbf{I}^{(z)} \right)$ comprises $3(m^{(z)} + 1)$ user-defined interpolation functions for $\ell = 0, 1, 2, \dots, m^{(z)}$ and $\mathbf{I}^{(z)}$ represents an identity matrix of the size of the number of degrees of freedom in subdomain $\Omega^{(z)}$. In a similar way, Lagrange multipliers $\lambda(t)$ may also be obtained at any time t by interpolation as:

$$\lambda(t) = w_0^\lambda(t) \lambda_0 + \sum_{\beta \in \mathcal{B}} w_\beta^\lambda(t) \lambda_\beta \quad (20)$$

where $w_0^\lambda(t)$ together with $w_\beta^\lambda(t)$ represent $p^I + 1$ user-defined interpolation functions. Note that λ_0 is known from initial conditions or from the previous block time step.

For interpolation, one may use different functions (such as Lagrange, Bézier, spline etc.), but one must ensure that they form a partition of unity i.e. $\sum_{\ell=0}^{m^{(z)}} w_\ell(t) = 1$. For example, Lagrange polynomials for interpolation subdomain states $\mathbf{x}$ may be defined as:

Fig. 4 Multi-time-step trajectories of subdomain state vectors $\mathbf{x}^{(z)}(t)$ and Lagrange multipliers $\lambda(t)$. Note that the subdomain state vectors $\mathbf{x}_\ell^{(z)}$ are only computed at $\Delta t^{(z)}$ intervals and Lagrange multipliers λ_β are only computed at times t_β . The exchange of information between subdomain $\Omega^{(z)}$ and the interface Γ^I at all other times t , is facilitated by interpolating these quantities with user-defined functions



$$L_\ell^{m^{(z)}}(t) = \prod_{\substack{i=0 \\ i \neq \ell}}^{m^{(z)}} \frac{t - t_i^{(z)}}{t_\ell^{(z)} - t_i^{(z)}} \quad \text{for } \ell = 0, 1, 2, \dots, m^{(z)}, \quad (21)$$

whereas Lagrange interpolating polynomials for the Lagrange multipliers λ may be defined as:

$$L_\beta^{p^I}(t) = \prod_{\substack{i \in \{0\} \cup \mathcal{B} \\ i \neq \beta}} \frac{t - t_i}{t_\beta - t_i} \quad \text{for } \beta \in \{0\} \cup \mathcal{B} \quad (22)$$

Examples of interpolating functions for $m^{(z)} = 1, 2$, and 3 are shown in Fig. 5.

Note that in Fig. 5, functions $w(t)$ are shown as interpolating functions. However, this is not a necessary requirement as one may use alternative approximation functions for $w(t)$ as well. Regardless, the choice of approximation functions can affect the stability and accuracy characteristics of the resulting MTS method. In this article, we restrict our attention to piecewise-linear and quadratic Lagrange polynomials and show that this choice works well for the problems presented in Sect. 4. A more detailed analysis of different approximation functions is left for the future.

3.2 Solution of subdomain state vectors

The first step of computing the MTS solution is to advance the state vectors for all subdomains. For a typical subdomain $\Omega^{(z)}$, the state vectors are advanced $m^{(z)}$ times by a time step of $\Delta t^{(z)}$ using the GRKN scheme of choice. To do this, the time-discrete governing equations of motion for subdomain $\Omega^{(z)}$ are obtained by modifying Eq. (4) to include the effect of the Lagrange multipliers:

$$M^{(z)} \bar{a}_{\ell+\gamma_i}^{(z)} + D_{\ell+\gamma_i}^{(z)} \bar{v}_{\ell+\gamma_i}^{(z)} + K_{\ell+\gamma_i}^{(z)} \bar{d}_{\ell+\gamma_i}^{(z)} + C^{(z)T} \lambda \left(t_{\ell+\gamma_i}^{(z)} \right) = f^{(z)} \left(t_{\ell+\gamma_i}^{(z)} \right) \quad (23)$$

Note that, since Eq. (23) is enforced at times $t_{\ell+\gamma_i}^{(z)}$ where $\ell = 0, 1, 2, \dots, (m^{(z)} - 1)$ and $i = 1, 2, \dots, p^{(z)}$ (shown with red dashed circles in Fig. 4), the Lagrange multipliers at these instants must be interpolated from their discrete values at t_β . Next, similar to Eqs. (8)–(10), algorithmic quantities $\bar{a}_{\ell+\gamma_i}^{(z)}$, $\bar{v}_{\ell+\gamma_i}^{(z)}$, and $\bar{d}_{\ell+\gamma_i}^{(z)}$ are expressed in terms of the state vectors $a_{\ell+\gamma_i}^{(z)}$, $v_{\ell+1}^{(z)}$, and $d_{\ell+1}^{(z)}$ as:

$$\sum_{j=0}^{p^{(z)}} M_{ij}^{(z)} a_{\ell+\gamma_j}^{(z)} = f^{(z)} \left(t_{\ell+\gamma_i}^{(z)} \right) - C^{(z)T} \lambda \left(t_{\ell+\gamma_i}^{(z)} \right) - \left(N_i^{(z)} v_\ell^{(z)} + K_{\ell+\gamma_i}^{(z)} d_\ell^{(z)} \right) \quad (24)$$

$$v_{\ell+1}^{(z)} = v_\ell^{(z)} + \Delta t^{(z)} \sum_{j=0}^{p^{(z)}} \nu_j^{(z)} a_{\ell+\gamma_j}^{(z)} \quad (25)$$

$$d_{\ell+1}^{(z)} = d_\ell^{(z)} + \delta_p^{(z)} \Delta t^{(z)} v_\ell^{(z)} + \left(\Delta t^{(z)} \right)^2 \sum_{j=0}^{p^{(z)}} \mu_j^{(z)} a_{\ell+\gamma_j}^{(z)} \quad (26)$$

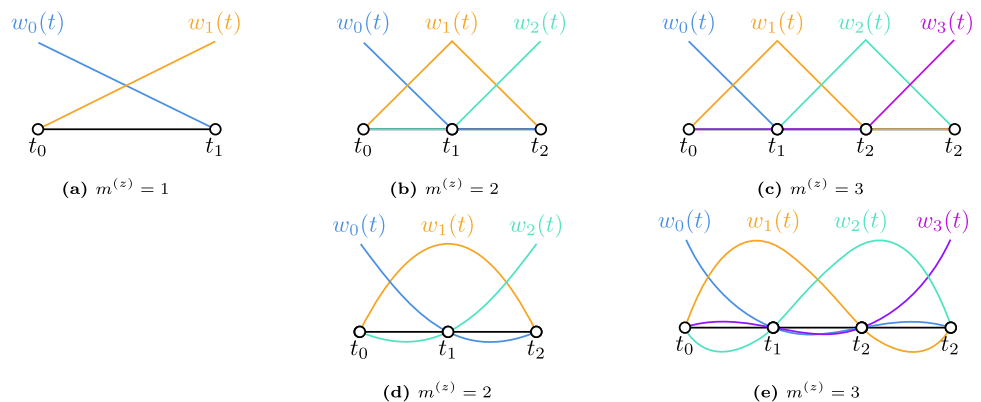
Substituting for $\lambda(t_{\ell+\gamma_i}^{(z)})$ from Eq. (20) and collecting Eq. (24) for $i = 1, 2, \dots, p^{(z)}$ in a manner similar to Eq. (11), we obtain:

$$\mathbb{M}^{(z)} a_{\ell+1}^{(z)} = \mathbb{f}_{\ell \rightarrow \ell+1}^{(z)} - \mathbb{C}^{(z)T} \left(\lambda_0 + \mathbb{W}_\beta^{(z)} \lambda_\beta \right) - \mathbb{N}^{(z)} x_\ell^{(z)} \quad (27)$$

where

$$\lambda_0 = \begin{bmatrix} w_0^\lambda(t_{\ell+\gamma_1}^{(z)}) \lambda_0 \\ w_0^\lambda(t_{\ell+\gamma_2}^{(z)}) \lambda_0 \\ \vdots \\ w_0^\lambda(t_{\ell+\gamma_p}^{(z)}) \lambda_0 \end{bmatrix}_{p^{(z)} \times 1}, \quad \lambda_\beta = \begin{bmatrix} \lambda_{\beta_1} \\ \lambda_{\beta_2} \\ \vdots \\ \lambda_{\beta_{p^I}} \end{bmatrix}_{p^I \times 1}, \quad \mathbb{C}^{(z)} = \begin{bmatrix} C^{(z)} & 0 & \dots & 0 \\ 0 & C^{(z)} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & C^{(z)} \end{bmatrix}_{p^{(z)} \times p^{(z)}} \quad (28)$$

Fig. 5 Examples of weight functions used for interpolation: **a–c** piecewise-linear interpolation; and **a, d, e** Lagrange polynomial



and the matrix of weights $\mathbb{W}_\beta^{(z)}$ is given by:

$$\mathbb{W}_\beta^{(z)} = \begin{bmatrix} w_{\beta_1}^\lambda \left(t_{\ell+\bar{\gamma}_1}^{(z)} \right) \mathbf{I}^I & w_{\beta_2}^\lambda \left(t_{\ell+\bar{\gamma}_1}^{(z)} \right) \mathbf{I}^I & \cdots & w_{\beta_{p^I}}^\lambda \left(t_{\ell+\bar{\gamma}_1}^{(z)} \right) \mathbf{I}^I \\ w_{\beta_1}^\lambda \left(t_{\ell+\bar{\gamma}_2}^{(z)} \right) \mathbf{I}^I & w_{\beta_2}^\lambda \left(t_{\ell+\bar{\gamma}_2}^{(z)} \right) \mathbf{I}^I & \cdots & w_{\beta_{p^I}}^\lambda \left(t_{\ell+\bar{\gamma}_2}^{(z)} \right) \mathbf{I}^I \\ \vdots & \vdots & \ddots & \vdots \\ w_{\beta_1}^\lambda \left(t_{\ell+\bar{\gamma}_p}^{(z)} \right) \mathbf{I}^I & w_{\beta_2}^\lambda \left(t_{\ell+\bar{\gamma}_p}^{(z)} \right) \mathbf{I}^I & \cdots & w_{\beta_{p^I}}^\lambda \left(t_{\ell+\bar{\gamma}_p}^{(z)} \right) \mathbf{I}^I \end{bmatrix}_{p^{(z)} \times p^I} \quad (29)$$

where $\mathbf{I}^I$ is the identity matrix of the size of the number of degrees of freedom on interface Γ^I .

In a manner similar to Eq. (13), Eqs. (25) and (26) may be combined with Eq. (27) to obtain:

$$\mathbb{x}_{\ell+1}^{(z)} = \mathbb{D}^{(z)} \mathbb{x}_\ell^{(z)} + \mathbb{P}^{(z)} \mathbb{a}_{\ell+1}^{(z)} \quad (30)$$

where the quantities $\mathbb{x}_\ell^{(z)}$, $\mathbb{x}_{\ell+1}^{(z)}$, $\mathbb{D}^{(z)}$, $\mathbb{P}^{(z)}$ and $\mathbb{a}_{\ell+1}^{(z)}$ are defined analogously to Eqs. (12) and (14). Finally, analogous to Eq. (15), combining Eq. (30) with Eq. (27), subdomain $\Omega^{(z)}$ can be advanced $m^{(z)}$ times by $\Delta t^{(z)}$ time steps using the following equation:

$$\mathbb{x}_{\ell+1}^{(z)} = \mathbb{G}^{(z)} \mathbb{x}_\ell^{(z)} + \mathbb{L}^{(z)} \left(\mathbb{f}_{\ell \rightarrow \ell+1}^{(z)} - \mathbb{C}^{(z)T} \boldsymbol{\lambda}_0 - \mathbb{C}^{(z)T} \mathbb{W}_\beta^{(z)} \boldsymbol{\lambda}_\beta \right)$$

freedom on the interface Γ^I . For instance, the $\mathbb{R}_\beta^{(z)}$ matrix corresponding to different types of continuity constraints may be defined as follows:

- Displacement continuity: $\mathbb{R}_\beta^{(z)} \equiv [\mathbf{0}, \mathbf{0}, \mathbf{C}^{(z)}]$
- Velocity continuity: $\mathbb{R}_\beta^{(z)} \equiv [\mathbf{0}, \mathbf{C}^{(z)}, \mathbf{0}]$
- Acceleration continuity: $\mathbb{R}_\beta^{(z)} \equiv [\mathbf{C}^{(z)}, \mathbf{0}, \mathbf{0}]$

Note that most existing MTS methods employ the velocity continuity constraint at time $t = t_0 + \Delta T$ since this constraint leads to stable solutions [46, 47, 54–56, 67, 68]. The proposed

$$\text{for } \ell = 0, 1, \dots, m^{(z)} - 1 \quad (31)$$

where $\mathbb{G}^{(z)} = \mathbb{D}^{(z)} - \mathbb{L}^{(z)} \mathbb{N}^{(z)}$ and $\mathbb{L}^{(z)} = \mathbb{P}^{(z)} \mathbb{M}^{(z)^{-1}}$. Note that in order to compute the state vectors $\mathbb{x}_\ell^{(z)}$ for all time steps from t_0 to $t_0 + \Delta T$, one needs to compute all the Lagrange multipliers $\boldsymbol{\lambda}_\beta$ over the block time step ΔT as well. This is done by enforcing multiple continuity constraints simultaneously as described next.

3.3 Computation of Lagrange multipliers with multiple simultaneous continuity constraints

In general, one may choose to impose p^I continuity constraints at different instants of time $t_{\bar{\beta}}$ as shown with blue dashed squares in Fig. 4. Note that the values of $\bar{\beta} \in \bar{\mathcal{B}}$ need not be identical to $\beta \in \mathcal{B}$ which denote the times at which Lagrange multipliers are discretized, shown in solid blue squares in Fig. 4. Further, values in the set $\bar{\mathcal{B}}$ need not be distinct either. Continuity constraints enforced at times $t_{\bar{\beta}}$ for all $\bar{\beta} \in \bar{\mathcal{B}}$ may be written as:

$$\sum_{z=1}^S \mathbb{R}_\beta^{(z)} \mathbb{x}_\ell^{(z)}(t_{\bar{\beta}}) = 0 \quad \text{for all } \bar{\beta} \in \bar{\mathcal{B}} \quad (32)$$

where the matrix $\mathbb{R}_\beta^{(z)}$ denotes a restriction operator that maps displacement, velocity, and acceleration degrees of freedom from subdomain $\Omega^{(z)}$ to constraint degrees of

MTS framework, however, is more general and encompasses the velocity continuity as a special case for the following parametric choices: $p^I = 1$, $\bar{\beta} = 1$, and $\mathbb{R}_1^{(z)} = [\mathbf{0}, \mathbf{C}^{(z)}, \mathbf{0}]$.

For general coupling constraints, the state vector $\mathbb{x}^{(z)}(t_{\bar{\beta}})$ in Eq. (32) is obtained by interpolation between its discrete values $\mathbb{x}_\ell^{(z)}$ using Eq. (18) as shown in Fig. 4. Substituting for $\mathbb{x}^{(z)}(t_{\bar{\beta}})$ from Eq. (18), the constraint equations may be expressed as:

$$\sum_{z=1}^S \mathbb{R}_\beta^{(z)} \left(\sum_{\ell=0}^{m^{(z)}} \mathbb{W}_\ell^{(z)}(t_{\bar{\beta}}) \mathbb{x}_\ell^{(z)} \right) = 0 \quad \text{for all } \bar{\beta} \in \bar{\mathcal{B}} \quad (33)$$

Combining all p^I constraint equations for all values of $\bar{\beta} \in \bar{\mathcal{B}}$, we obtain:

$$\sum_{z=1}^S \sum_{\ell=0}^{m^{(z)}} \mathbb{B}_\ell^{(z)} \mathbb{x}_\ell^{(z)} = 0 \quad (34)$$

where $\mathbb{B}_\ell^{(z)} = \left[\mathbb{R}_{\beta_1}^{(z)} \mathbb{W}_\ell^{(z)}(t_{\bar{\beta}_1}); \mathbb{R}_{\beta_2}^{(z)} \mathbb{W}_\ell^{(z)}(t_{\bar{\beta}_2}); \dots \right]$ is a $p^I \times 1$ block matrix where each block is of the size of the interface restriction operator $\mathbb{R}_\beta^{(z)}$. Eq. (34) represents the global constraint equation that enforces continuity of all the subdomain solutions across the interface Γ^I .

Finally, to advance the entire problem domain by one block time step ΔT , Eqs. (31) and (34) need to be solved simultaneously. To do so, we substitute Eq. (31) into Eq. (34) leading to the following algebraic equations for subdomains $\Omega^{(z)}$:

$$\left(\sum_{z=1}^S \sum_{\ell=0}^{m^{(z)}} \mathbb{B}_{\ell}^{(z)} \mathbb{Y}_{\ell}^{(z)} \right) \boldsymbol{\lambda}_{\beta} = \left(\sum_{z=1}^S \sum_{\ell=0}^{m^{(z)}} \mathbb{B}_{\ell}^{(z)} \mathbb{u}_{\ell}^{(z)} \right) \quad (35)$$

where state vector $\mathbb{u}_{\ell+1}^{(z)}$ in each subdomain are computed first by ignoring the contribution of Lagrange multipliers:

$$\mathbb{u}_{\ell+1}^{(z)} = \mathbb{G}^{(z)} \mathbb{u}_{\ell}^{(z)} + \mathbb{L}^{(z)} \left(\mathbb{f}_{\ell \rightarrow \ell+1}^{(z)} - \mathbb{C}^{(z)T} \boldsymbol{\lambda}_0 \right) \quad \text{for } \ell = 0, 1, \dots, m^{(z)} - 1 \quad (36)$$

noting that $\mathbb{u}_0^{(z)} = \mathbb{x}_0^{(z)}$. The block matrices $\mathbb{Y}$ are obtained as:

$$\mathbb{Y}_{\ell+1}^{(z)} = \mathbb{G}^{(z)} \mathbb{Y}_{\ell}^{(z)} + \mathbb{L}^{(z)} \mathbb{C}^{(z)T} \mathbb{W}_{\beta}^{(z)} \quad \text{for } \ell = 0, 1, \dots, m^{(z)} - 1 \quad (37)$$

with $\mathbb{Y}_0^{(z)} = 0$. Then, from Lagrange multipliers $\boldsymbol{\lambda}_{\beta}$ in Eq. (35), we correct the subdomain solutions to incorporate the effect of the Lagrange multipliers as:

$$\mathbb{x}_{\ell+1}^{(z)} = \mathbb{u}_{\ell+1}^{(z)} - \mathbb{Y}_{\ell+1}^{(z)} \boldsymbol{\lambda}_{\beta} \quad (38)$$

Thus, we solve for the state vectors $\mathbb{u}_{\ell+1}^{(z)}$ from Eq. (36) first, then compute Lagrange multipliers from Eq. (35), and correct the subdomain solutions using Eq. (38). Note that there are no iterations required for coupling subdomains together.

To facilitate the formulation of new MTS methods using the proposed MTS framework, we present the Butcher tableau for MTS methods in Table 4. Note that several existing MTS methods in the literature are obtained as a special case of the proposed MTS framework as shown in Table 4.

3.4 Stabilization using a Scalar Auxiliary Variable (SAV)

A key difficulty with formulating general MTS methods is ensuring the stability of the resulting algorithm. Thus, even though the MTS framework described in Sects. 3.2 and 3.3 allows one to design general MTS methods with any number of constraint equations simultaneously, the resulting algorithm may not be stable. To ensure unconditional stability of the MTS algorithm, we adopt an approach described by Shen and co-workers [71, 72] that uses a scalar auxiliary

Table 4 Butcher tableau for MTS methods

MTS (Underlying time integration)	p^I	Governing equations (related to $\lambda(t)$)		Constraint conditions (related to $\mathbb{x}^{(z)}(t)$)		
Existing MTS methods		$\mathcal{B}$	$w_0^\lambda(t)$	$w_\beta^\lambda(t)$	$\overline{\mathcal{B}}$	$\mathbb{R}_\beta^{(z)}$
VC-Newmark/PH* [46] (Newmark [5])	1	$\{1\}$	$L_0^1(t)$	$L_1^1(t)$	$\{1\}$	$[\mathbf{0}, C^{(z)}, \mathbf{0}]$
Karimi & Nakshatrala [49] (Newmark [5])	1	$\{1\}$	$L_0^1(t)$	$L_1^1(t)$	$\{1\}$	$[\mathbf{0}, C^{(z)}, \mathbf{0}]$
Mahjoubi & Krenk [52] (Krenk [69])	1	$\{0.5\}$	0	1	$\{1\}$	$[\mathbf{0}, \mathbf{0}, C^{(z)}]$
BGC-macro [54] (HHT- α [8], WBZ- α [9], CH- α [10])	1	$\{1\}$	$L_0^1(t)$	$L_1^1(t)$	$\{1\}$	$[\mathbf{0}, C^{(z)}, \mathbf{0}]$
DAE-GS4 [56] (GS4 [14])	1	$\{1\}$	$L_0^1(t)$	$L_1^1(t)$	$\{1\}$	$[\mathbf{0}, C^{(z)}, \mathbf{0}]$
VC-Bathe* [70] (ρ_∞ -Bathe [20])	1	$\{1\}$	$L_0^1(t)$	$L_1^1(t)$	$\{1\}$	$[\mathbf{0}, C^{(z)}, \mathbf{0}]$
Proposed MTS methods		$\mathcal{B}$	$w_0^\lambda(t)$	$w_\beta^\lambda(t)$	$\overline{\mathcal{B}}$	$\mathbb{R}_\beta^{(z)}$
VC-GRKN (All implicit and explicit GRKN schemes)	1	$\{1\}$	$L_0^1(t)$	$L_1^1(t)$	$\{1\}$	$[\mathbf{0}, C^{(z)}, \mathbf{0}]$
VCDC-1 (Newmark [5], CH- α [10], GS4 [14] ρ_∞ -Bathe [20], Noh-Bathe [18])	2	$\{0.5, 1\}$	Piecewise-linear interp**		$\{1, 1\}$	$\{[\mathbf{0}, C^{(z)}, \mathbf{0}], [\mathbf{0}, \mathbf{0}, C^{(z)}]\}$
VCDC-2 (Newmark [5], CH- α [10], GS4 [14] ρ_∞ -Bathe [20], Noh-Bathe [18])	2	$\{0.5, 1\}$	$L_0^2(t)$	$L_\beta^2(t)$	$\{1, 1\}$	$\{[\mathbf{0}, C^{(z)}, \mathbf{0}], [\mathbf{0}, \mathbf{0}, C^{(z)}]\}$

*Includes a correction term for the Lagrange multipliers $\boldsymbol{\lambda}(t)$ (see Appendix B for details)

** For piecewise-linear interpolation, see Fig. 5b

variable (SAV) to update the solution in Eq. (38). In this section, we show that the SAV approach guarantees that any MTS method designed using the proposed framework will result in an unconditionally energy stable algorithm.

Note that accuracy of the resulting MTS method will depend upon the chosen constraint equations and the accuracy of the underlying time integration methods used in each subdomain. Thus, we do not make any claims about the accuracy of the resulting MTS algorithm, but we do assume that the time integration method and the constraints chosen by the user result in a *consistent* time-discretization. Accuracy of the resulting MTS algorithm will need to be studied individually for each use-case of proposed MTS framework.

For stabilization, we first define a SAV, r , and its time derivative, $\dot{r}$, for the entire problem domain as:

$$r(t) = \sum_{z=1}^S E^{(z)} [\mathbf{u}^{(z)}(t), \dot{\mathbf{u}}^{(z)}(t)] + c_0 \quad \text{with } c_0 > 0 \quad (39)$$

$$\frac{\partial r(t)}{\partial t} = \frac{\partial \left(\sum_{z=1}^S E^{(z)} + c_0 \right)}{\partial t} = \sum_{z=1}^S G^{(z)} \quad (40)$$

where $E^{(z)}$ is a measure of the energy in each of the subdomains $\Omega^{(z)}$, $G^{(z)}$ is the time rate of change of $E^{(z)}$, and c_0 is a positive user-defined constant. For instance, in linear elasto-dynamics, $E^{(z)}$ and $G^{(z)}$ may be defined, respectively, as:

$$E^{(z)} [\mathbf{u}^{(z)}, \dot{\mathbf{u}}^{(z)}] = \frac{1}{2} \left(\dot{\mathbf{u}}^{(z)T} \mathbf{M}^{(z)} \dot{\mathbf{u}}^{(z)} + \mathbf{u}^{(z)T} \mathbf{K}^{(z)} \mathbf{u}^{(z)} \right), \quad (41)$$

$$G^{(z)} [\dot{\mathbf{u}}^{(z)}] = \dot{\mathbf{u}}^{(z)T} \left(\mathbf{f}^{(z)} - \mathbf{D}^{(z)} \dot{\mathbf{u}}^{(z)} \right) \quad (42)$$

Note that since $\mathbf{M}^{(z)}$ is symmetric and positive-definite, and $\mathbf{D}^{(z)}$ and $\mathbf{K}^{(z)}$ are semi-positive-definite and symmetric, $E^{(z)} \geq 0$ and $G^{(z)} \leq 0$ when $\mathbf{f}^{(z)} = \mathbf{0}$. The value of c_0 may be chosen as any positive constant, large or small. However, as a guideline (see Sect. 4.1), we recommend using $c_0 = 10^4 E_{max}$, where E_{max} is an estimate of maximum energy of the entire system over the entire duration of the simulation $[0, T]$.

We time-discretize the SAV in Eq. (39) and its evolution Eq. (40) and solve them together with Eqs. (31) and (34) to advance the SAV from t_0 to $t_0 + \Delta T$ using the backward difference scheme:

$$\frac{r_1 - r_0}{\Delta T} = \frac{r_1}{\sum_{z=1}^S E^{(z)} [\mathbf{d}_m^{(z)}, \mathbf{v}_m^{(z)}] + c_0} \left(\sum_{z=1}^S G^{(z)} [\mathbf{v}_m^{(z)}] \right) \quad (43)$$

where $r(t_0) \approx r_0$ and $r(t_0 + \Delta T) \approx r_1$. We solve for r_1 from Eq. (43) and update the MTS coupled solution for each subdomain as:

$$\mathbf{x}_{\ell+1}^{(z)} = \eta_{\ell+1}^{(z)} \mathbf{x}_{\ell+1}^{(z)} \quad \text{for } \ell = 0, 1, \dots, m^{(z)} - 1 \quad (44)$$

where the stabilization factor $\eta_{\ell+1}^{(z)}$ is obtained from the following sequence of computations:

$$\xi_m = \frac{r_1}{\sum_{z=1}^S E^{(z)} [\mathbf{d}_m^{(z)}, \mathbf{v}_m^{(z)}] + c_0} \quad (45)$$

$$\xi_{\ell+1}^{(z)} = \frac{m^{(z)} - (\ell + 1)}{m^{(z)}} \xi_0 + \frac{(\ell + 1)}{m^{(z)}} \xi_m \quad (46)$$

$$\eta_{\ell+1}^{(z)} = 1 - \left(1 - \xi_{\ell+1}^{(z)} \right)^3 \quad (47)$$

Finally, the stabilized MTS coupled solution $\mathbf{x}_0^{(z)}$ is used as the initial state vector for the next block time step ΔT i.e. for each subdomain $\mathbf{x}_0^{(z)} = \mathbf{x}_0^{\prime(z)}$. This process is repeated successively to advance the MTS coupled solution for the duration of the simulation.

3.5 Proof of unconditional stability

Here, we prove that the solutions generated using the proposed MTS framework and stabilized with the SAV approach lead to unconditionally stable MTS methods.

Theorem 3.1 *For a linear system in the absence of external forces, the proposed MTS framework leads to unconditional energy stability such that*

$$r_1 \leq r_0 \quad (48)$$

and there exists $U > 0$ such that

$$E^{(z)} [\mathbf{d}_m^{\prime(z)}, \mathbf{v}_m^{\prime(z)}] \leq U^2 \quad (49)$$

Proof From Eq. (41), $E \geq 0$ and $G \leq 0$ at all times for linear problems without external forces. Thus, given $r_0 > 0$, Eq. (43) ensures

$$r_1 = \frac{r_0}{1 - \Delta T \frac{\sum_{z=1}^S G^{(z)}[\mathbf{v}_m^{(z)}]}{\sum_{z=1}^S E^{(z)}[\mathbf{d}_m^{(z)}, \mathbf{v}_m^{(z)}] + c_0}} > 0 \quad (50)$$

Then, given $r_1 > 0$, Eq. (43) leads to

$$r_1 - r_0 \leq 0 \quad (51)$$

This proves Eq. (48).

Next, we prove Eq. (49). From Eqs. (45) and (48), we obtain

$$\xi_m = \frac{r_1}{\sum_{z=1}^S E^{(z)}[\mathbf{d}_m^{(z)}, \mathbf{v}_m^{(z)}] + c_0} \leq \frac{r_0}{\sum_{z=1}^S E^{(z)}[\mathbf{d}_m^{(z)}, \mathbf{v}_m^{(z)}] + c_0} \quad (52)$$

In addition, $\eta_m^{(z)}$ can be expressed as $\eta_m^{(z)} = \xi_m P(\xi_m)$, where $P(\xi_m)$ is a quadratic polynomial of ξ_m . Thus, there exist U such that $|P(\xi_m)| \leq 2\sqrt{c_0}U/r_0$. Therefore, from Eq. (52),

$$\begin{aligned} |\eta_m^{(z)}| &= |\xi_m P(\xi_m)| \leq \frac{2\sqrt{c_0}U}{\sum_{z=1}^S E^{(z)}[\mathbf{d}_m^{(z)}, \mathbf{v}_m^{(z)}] + c_0} \\ &\leq \frac{2\sqrt{c_0}U}{E^{(z)}[\mathbf{d}_m^{(z)}, \mathbf{v}_m^{(z)}] + c_0} \end{aligned} \quad (53)$$

From Eqs. (41) and (44),

$$\begin{aligned} E^{(z)}[\mathbf{d}_m^{(z)}, \mathbf{v}_m^{(z)}] &= \left(\eta_m^{(z)}\right)^2 E^{(z)}[\mathbf{d}_m^{(z)}, \mathbf{v}_m^{(z)}] \\ &\leq \left(\frac{2\sqrt{c_0}U}{E^{(z)}[\mathbf{d}_m^{(z)}, \mathbf{v}_m^{(z)}] + c_0}\right)^2 E^{(z)}[\mathbf{d}_m^{(z)}, \mathbf{v}_m^{(z)}] \leq U^2 \end{aligned} \quad (54)$$

The proof is complete. $\square$

Note that the SAV approach is able to achieve unconditional stability for general nonlinear problems as well [71, 72]. In particular, using arguments similar to [73], it can be shown that there exists $U > 0$ such that $E^{(z)}[\mathbf{d}_{nm}^{(z)}, \mathbf{v}_{nm}^{(z)}] \leq U^2 \quad \forall n \in [0, T/\Delta T]$. However, we leave a detailed proof of stability for nonlinear problems for a future effort.

3.6 Computational cost

In this section, we discuss the computational cost of MTS methods designed using the proposed MTS framework. This may be used to obtain an *a priori* estimate of the computational gain that can be achieved using different MTS parameters and to determine whether the MTS approach will be faster than a UTS approach for given problem.

First, we estimate the computational cost of a p -stage GRKN time integration scheme given in Eq. (15). Let the number of degrees of freedom in the problem domain be n and let the bandwidth of the domain matrices be l for each of the p sub-stages of the GRKN scheme. The primary computational cost of GRKN schemes results from solving the linear system $\mathbb{M}^{-1}\mathbf{f}$ from Eq. (11). However, since most time integration schemes fall into the category of diagonally implicit GRKN schemes for which $M_{ij} = 0$ for $j > i$, we restrict our attention to only to this class of GRKN schemes. In this case, we can compute the solution by solving the linear system of equations $M_{ii}^{-1}\mathbf{f}(t_{\bar{\gamma}_i})$ with $i = 1, 2, \dots, p$ for each of the p sub-stages. Since, the computational cost of solving a single sub-stage of a GRKN scheme is $O(l^2n)$ (cost of factorization of M_{ii}), the cost for a p -stage GRKN scheme may be estimated as $O(pl^2n)$. For nonlinear problems, this would be the computational cost associated with every iteration of every time step. Thus, the total computational cost of a uniform time-step (UTS) method for nonlinear problems may be estimated as:

$$C \geq c_P \frac{T}{\Delta t} i pl^2n \quad (55)$$

where i is the average number of nonlinear iterations per time step, and c_P is a constant that depends upon the hardware platform. On the other hand, for linear problems, factorization needs to be conducted only once and the cost for each time step reduces to that of a matrix-vector product i.e. $O(pln)$. Thus, the total computational cost of a UTS method for linear problems is:

$$C \geq \underbrace{c_P pl^2n}_{\approx 0} + c_P \frac{T}{\Delta t} pln \quad (56)$$

where the first term (related to initial factorization) may be neglected for long-time simulations i.e. when $T \gg \Delta t$. The UTS computational costs given in Eqs. (55) and (56) serve as the baseline for comparison with MTS methods.

To estimate the cost of a MTS method, let the problem domain be decomposed into S subdomains with $n^{(z)}$ degrees of freedom in subdomain $\Omega^{(z)}$ with bandwidth $l^{(z)}$ and let n^I be the number of degrees of freedom along the interface

Γ^I . The total computational cost of a MTS method can then found as the sum of the costs results from computing the five quantities as given in Table 5.

Note that the cost of computing the predictor $u_\ell^{(z)}$ from Eq. (36) is simply the cost of a $p^{(z)}$ -stage GRKN scheme applied to each subdomain times the time-step ratio $m^{(z)}$ and summed up for all the S subdomains. Next, the interface operator $\mathbb{Y}_\ell^{(z)}$ in Eq. (37) requires $n^I p^I$ matrix-vector operations (for each column of the $\mathbb{Y}_\ell^{(z)}$ matrix) with subdomain operator $M_{ii}^{(z)}$ that has already been factorized in the predictor stage. Then, computing Lagrange multipliers λ_β from Eq. (35) requires factorization of the interface operator $[\mathbb{B}_\ell^{(z)} \mathbb{Y}_\ell^{(z)}]$. Since the size of this operator is $n^I p^I$ and it is a full matrix, the associated cost is $O(n^I p^I)^3$. The correction $x_\ell^{(z)}$ in Eq. (38) involves a matrix-vector multiplication of size $n^I p^I$ for $\mathbb{Y}_\ell^{(z)} \lambda_\beta$ for every subdomain $\Omega^{(z)}$ and at each of the $m^{(z)}$ intermediate time steps. Finally, the stabilization in Eq. (44) is a scalar multiplication with negligible cost. Thus, over the duration of the simulation $[0, T]$, the total computational cost of MTS methods for nonlinear problems may be estimated as:

$$C \geq \frac{T}{\Delta T} (i(c_P C_P + c_I C_I + c_L C_L + c_C C_C) + c_S C_S) \quad (57)$$

where i is the average number of nonlinear iterations per time step, and c_P , c_I , c_L , c_C , and c_S are constants that primarily depend upon the hardware platform and the linear algebra library subroutines used for these computations. For linear problems, the matrix factorizations of the subdomain operators and the interface operator can be computed in advance of the time-stepping loop. Thus, the total computational cost of MTS methods for linear problems may be outlined as:

$$C \geq \underbrace{c_P C_P + c_I C_I + c_L C_L}_{\approx 0} + \frac{T}{\Delta T} (c_P C_P + c_L C_L + c_C C_C + c_S C_S) \quad (58)$$

where the first term (related to initial factorizations) may be neglected if $T \gg \Delta T$.

While the estimates provided above assume the use of a direct solver for all computational stages, large problems that use domain decomposition and MTS are usually solved with iterative solvers—especially on parallel computing platforms. Computational cost estimates with iterative solvers on parallel computing hardware depend heavily upon the implementation. For instance, Farhat and coworkers [42, 74] developed implementations where direct solvers are used to compute subdomain solutions whereas an iterative preconditioned conjugate gradient (PCG) solver is used to couple the subdomain solutions together. The computational cost estimates provided in Table 5 remain largely valid for such implementations, except that component C_I of the cost is eliminated for iterative solvers and the remaining computational cost is multiplied by the number of PCG iterations needed to compute the solution. Further, if one uses iterative solvers that scale well (i.e. reduce computational time as the number of processors is increased—see [75]), then the number of PCG iterations are usually independent of the problem-size and that leads to improved parallel performance. For MTS methods, scalability can be achieved by a careful decomposition of the problem domain into subdomains, accounting for the number of degrees of freedom and the time step for each subdomain so that the computational load can be balanced among all the processors efficiently. A recursive approach that achieves almost optimal MTS decompositions and leads to balanced computational loads on processors is presented in [76].

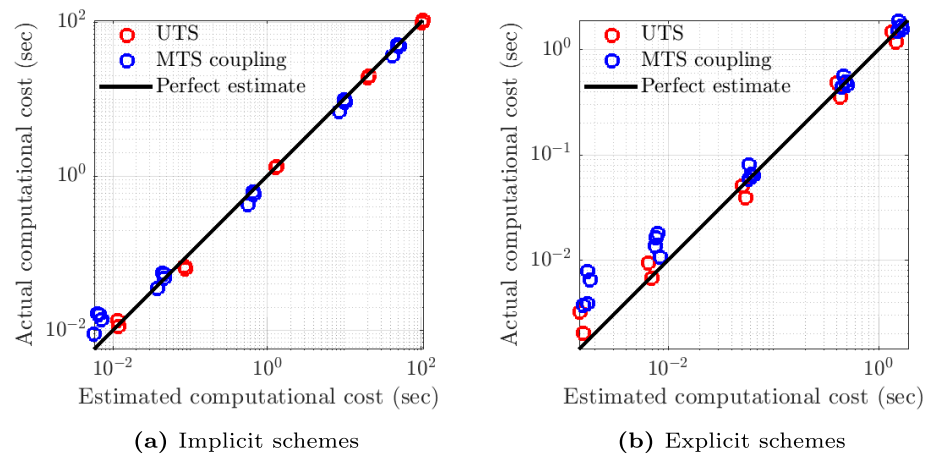
Figure 6 shows a comparison of the estimated computational cost from Table 5 and the actual computational cost for the example problem presented in Sect. 4.2 when solved with different UTS and MTS methods on a Linux desktop with Intel Xeon 2.4 GHz processor with 32 cores, 64 GB RAM, and 3 TB hard disk. The constants c_P , c_I , c_L , c_C , and c_S for this hardware platform are obtained by a least-squares fit of the estimated cost against actual runtimes. Even though the calibrated values of these constants are specific to this hardware platform, one can still use the computational cost analysis presented above to estimate potential computational gains of different MTS decompositions over a UTS approach (approximately) on other hardware platforms as

Table 5 Components of computational cost for MTS methods

	Cost to compute	Nonlinear	Linear
C_P :	Predictor $u_\ell^{(z)}$	$O^* \left(\sum_{z=1}^S m^{(z)} p^{(z)} \Gamma^{(z)^2} n^{(z)} \right)$	$O \left(\sum_{z=1}^S m^{(z)} p^{(z)} \Gamma^{(z)} n^{(z)} \right)$
C_I :	Interface operator $\mathbb{Y}_\ell^{(z)}$	$O^* \left(n^I p^I \sum_{z=1}^S m^{(z)} p^{(z)} \Gamma^{(z)} n^{(z)} \right)$	
C_L :	Lagrange multiplier λ_β	$O^* (n^I p^I)^3$	$O(n^I p^I)^2$
C_C :	Correction $x_\ell^{(z)}$	$O \left(\sum_{z=1}^S m^{(z)} n^{(z)} n^I p^I \right)$	$O \left(\sum_{z=1}^S m^{(z)} n^{(z)} n^I p^I \right)$
C_S :	Stabilization $x_\ell'^{(z)}$	$O \left(\sum_{z=1}^S m^{(z)} n^{(z)} \right)$	$O \left(\sum_{z=1}^S m^{(z)} n^{(z)} \right)$

*For linear problems, these operators are carried out only once in advance of the time-stepping loop

Fig. 6 Comparison of estimated and actual computational costs for UTS and MTS methods



well. Such comparisons may be helpful in deciding, for instance, how many subdomains should be used, how many elements should each subdomain have, and what time step and numerical scheme should each subdomain be integrated with, irrespective of the hardware platform.

To compare the computational cost of a UTS scheme to that of a MTS method, we assume that both approaches will result in similar levels of accuracy if the time step Δt of the UTS method is identical to the minimum time step $\Delta t^{(z)}$ among all subdomains $\Omega^{(z)}$ of the MTS method (i.e. $\Delta t = \min(\Delta t^{(z)})$). This assumption is based on prior experience and observations with existing MTS methods. Problems in which a small region of a large spatial domain requires a finer resolution than the rest of the domain, such as the example presented in Sect. 4.3, are well-suited for the MTS approach because it allows one to define a small subdomain around the region of interest that is integrated with Δt while integrating other subdomains with larger time steps $\Delta t^{(z)} > \Delta t$. Further, if the number of Lagrange multipliers $n^I p^I$ is smaller than the average number of degrees of freedom per subdomain, then the computational cost of a MTS method is dominated by C_P :

$$C \approx c_P \sum_{z=1}^S \frac{T}{\Delta t^{(z)}} p^{(z)} l^{(z)^2} n^{(z)} \quad (59)$$

In contrast, with the UTS approach, one would have to use a small time step Δt for the entire problem domain, for which the computational cost is given by Eq. (55).

Comparing the two costs for the same p -stage GRKN scheme, we note that for most temporal multi-scale problems, the MTS cost $O(\sum_{z=1}^S l^{(z)^2} n^{(z)} / \Delta t^{(z)})$ will be less than the underlying UTS cost $O(l^2 n / \Delta t)$, thereby showing that MTS methods, in general, are more computationally efficient than UTS methods for comparable accuracy. In particular, for two subdomains $\Omega^{(1)}$ and $\Omega^{(2)}$ integrated with

ΔT and $\Delta t = \Delta T/m$, respectively, such that the region of interest $\Omega^{(2)}$ is much smaller than $\Omega^{(1)}$ (i.e. $n^{(2)} \ll n^{(1)}$), the computational cost of a MTS method is approximately m times smaller than that of the underlying UTS schemes.

4 Numerical examples

In this section, we demonstrate use-cases of the proposed MTS framework by designing new MTS methods for a variety of different time integration schemes. First, we consider the split single-degree-of-freedom (SDOF) problem [46] that serves as a critical benchmark for evaluating the performance of MTS methods. Next, we solve three problems, (1) wave propagation of a scalar field in a multi-material medium, (2) Lamb's problem with multiple spatial scales, and (3) impact problem of a sandwich plate. We show that existing MTS methods that are based on continuity of velocities lead to large drifts in displacements, whereas MTS methods designed using the proposed framework to enforce continuity of displacements and velocities are able to achieve high accuracy and unconditional stability at a low computational cost.

4.1 Split single-degree-of-freedom problem

The split single-degree-of-freedom (SDOF) problem serves as a benchmark for evaluating MTS methods (see [46] for details). As shown in Fig. 7, a SDOF oscillator is split into two SDOF oscillators coupled together by Lagrange multipliers that enforce certain continuity constraints. The split-SDOF oscillators are treated as subdomains and are integrated with different time integration schemes and different time steps. To test the performance of different MTS methods, results from the split-SDOF system are compared to the exact solution of the underlying combined SDOF system. Note that using modal decomposition, a linear

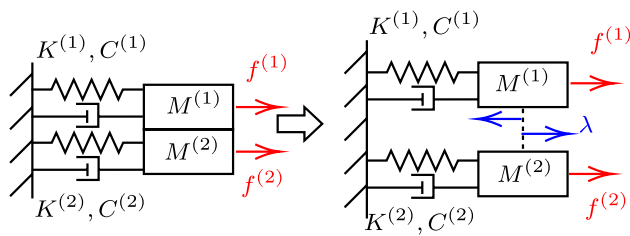


Fig. 7 Split single-degree-of-freedom (SDOF) problem with two subdomains

multi-degree-of-freedom (MDOF) system may be reduced to multiple, independent SDOF systems where the mass, damping, and stiffness for each SDOF system is a scalar value. Scaling the equation of motion to have unit mass, we obtain: $M^{(z)} \rightarrow 1$; $D^{(z)} \rightarrow 2\zeta^{(z)}\omega_0^{(z)}$; $K^{(z)} \rightarrow \omega_0^{(z)^2}$; and $f^{(z)} \rightarrow r^{(z)}$, where $\omega_0^{(z)}$, $\zeta^{(z)}$, and $r^{(z)}$ are the modal natural frequency, damping ratio, and generalized force, respectively, in subdomain $\Omega^{(z)}$. Without loss of generality, we write the equations of motion for a split-SDOF system with two SDOF subdomains $\Omega^{(1)}$ and $\Omega^{(2)}$:

$$M^{(z)}\ddot{u}^{(z)}(t) + 2M^{(z)}\zeta^{(z)}\omega_0^{(z)}\dot{u}^{(z)}(t) + M^{(z)}\left(\omega_0^{(z)}\right)^2 u^{(z)}(t) + C^{(z)T}\lambda(t) = M^{(z)}r^{(z)}(t) \quad \text{for } z = 1, 2 \quad (60)$$

where $C^{(1)} = 1$ and $C^{(2)} = -1$. The underlying combined SDOF system corresponding to Eq. (60) is given by:

$$(1 + \delta_M)\ddot{u}(t) + 2\zeta^{(1)}\omega_0^{(1)}(1 + \delta_M\delta_\zeta\delta_\omega)\dot{u}(t) + \left(\omega_0^{(1)}\right)^2(1 + \delta_M\delta_\omega^2)u(t) = r^{(1)}(t) + \delta_M r^{(2)}(t) \quad (61)$$

where $\delta_M = M^{(2)}/M^{(1)}$, $\delta_\zeta = \zeta^{(2)}/\zeta^{(1)}$, and $\delta_\omega = \omega_0^{(2)}/\omega_0^{(1)}$. We select $M^{(1)} = 1$, $\zeta^{(1)} = 0.02$, $\omega_0^{(1)} = 2\pi$, $\delta_M = 2$, $\delta_\zeta = 1.5$, $\delta_\omega = 6$, and $r^{(1)} = r^{(2)} = 0$ to represent a stiff-flexible coupled system under free vibration with a nonzero initial displacement and velocity: $d_0 = v_0 = 1$.

In Fig. 8, the split-SDOF system is solved with different existing and proposed new MTS methods. We denote the different MTS methods presented as: $\langle \text{MTS}_{\text{name}} \rangle (\Delta T)$: GRKN($m^{(1)}$)-GRKN($m^{(2)}$). The first column of Fig. 8 includes four existing MTS methods namely, VC-Newmark/PH method [46], BGC-macro method for CH- α schemes [54], DAE-GS4 method for GS4 schemes [56], and VC-Bathe method for ρ_∞ -Bathe schemes [70]. In the second and third columns of Fig. 8 we present two new classes of proposed MTS methods that simultaneously impose velocity and displacement continuity (VCDC) (i.e. $p^I = 2$) at every block time step ΔT (i.e. $\bar{\mathcal{B}} = \{1, 1\}$ and $\mathcal{B} = \{0.5, 1\}$). The first class of methods (VCDC-1) uses piecewise-linear interpolation functions $w(t)$ for

Lagrange multipliers and kinematic quantities, whereas the second class of methods (VCDC-2) uses a quadratic interpolation, as listed in Table 4. Note that for VCDC-1 and VCDC-2 MTS methods, the velocity and displacement constraints are both enforced at $t_{\bar{\mathcal{B}}} = t_0 + \Delta T$ where state vectors for both subdomains $\mathbf{x}_m^{(z)}$ are readily available. Thus, interpolation functions $\mathbb{W}_\ell^{(z)}(t)$ for subdomain state vectors are not needed i.e. any interpolation function may be used.

Duration of the simulation is $T = 1$ and the block time step for the first three existing MTS methods (VC-Newmark, BGC-macro, and DAE) is chosen to be $\Delta T = \Delta t^{(1)} = 0.05$ with time-step ratios of $m^{(1)} = 1$ and $m^{(2)} = 6$ resulting in $\Delta t^{(2)} = 1/120 \approx 0.0083$. For the VC-Bathe method, we use the same time-step ratios $m^{(1)} = 1$ and $m^{(2)} = 6$, but with $\Delta T = \Delta t^{(1)} = 0.1$ because the underlying time integration method, ρ_∞ -Bathe, is a 2-stage composite scheme and thus for time-step parity we use ΔT that is twice that of the single-stage time integration schemes. For the new proposed VCDC MTS methods, we use a block time step $\Delta T = 0.1$ and time-step ratios $m^{(1)} = 2$ and $m^{(2)} = 12$ for the single-stage schemes such as TR, CH- α , and GS4-U0V0 (subfigures (b), (c), (e), (f), (h), and (i)). This is because we found that enforcing the VCDC constraint with $m^{(z)} = 1$ for single-stage schemes leads to high errors. Instead, using time-step ratios $m^{(z)} \geq 2$ for all subdomains integrated with single-stage schemes gives stable and accurate solutions. For all MTS methods, we choose the same algorithmic parameters in both SDOF subdomains: trapezoidal rule (TR) with Newmark parameters $(\gamma, \beta) = (0.5, 0.25)$; CH- α scheme with $\rho_\infty = 0$; U0V0 scheme with $(\rho_\infty^{\min}, \rho_\infty^{\max}, \rho_\infty^s) = (0, 1, 0)$; and the ρ_∞ -Bathe scheme with $(\rho_\infty, \gamma) = (1 - \sqrt{3}, 1 + 1/\sqrt{3})$.

Note that the natural period T_p of the underlying combined SDOF system is 0.2027 and the block time step ΔT chosen here is much larger than what one would typically use to obtain a good resolution. This is done to highlight the drift between subdomain solutions resulting from the various MTS methods. We define drift between subdomains $\Omega^{(1)}$ and $\Omega^{(2)}$ at every block time step ΔT as:

$$D_x = \frac{\max_{n=0,1,\dots,N} |x_{nm}^{(1)} - x_{nm}^{(2)}|}{\max_{t \in [0,T]} \tilde{x}(t) - \min_{t \in [0,T]} \tilde{x}(t)} \quad (62)$$

where x is the quantity of interest (here displacement) and $\tilde{x}$ is its analytical solution. The number of block time steps over the duration of the simulation is $N (= \lfloor T/\Delta T \rfloor)$ where $\lfloor \cdot \rfloor$ denotes the nearest integer function. Figure 8 shows that existing MTS methods lead to large drifts in displacement because they are based only on continuity of velocities (i.e. $p^I = 1$), whereas both classes of the proposed VCDC MTS

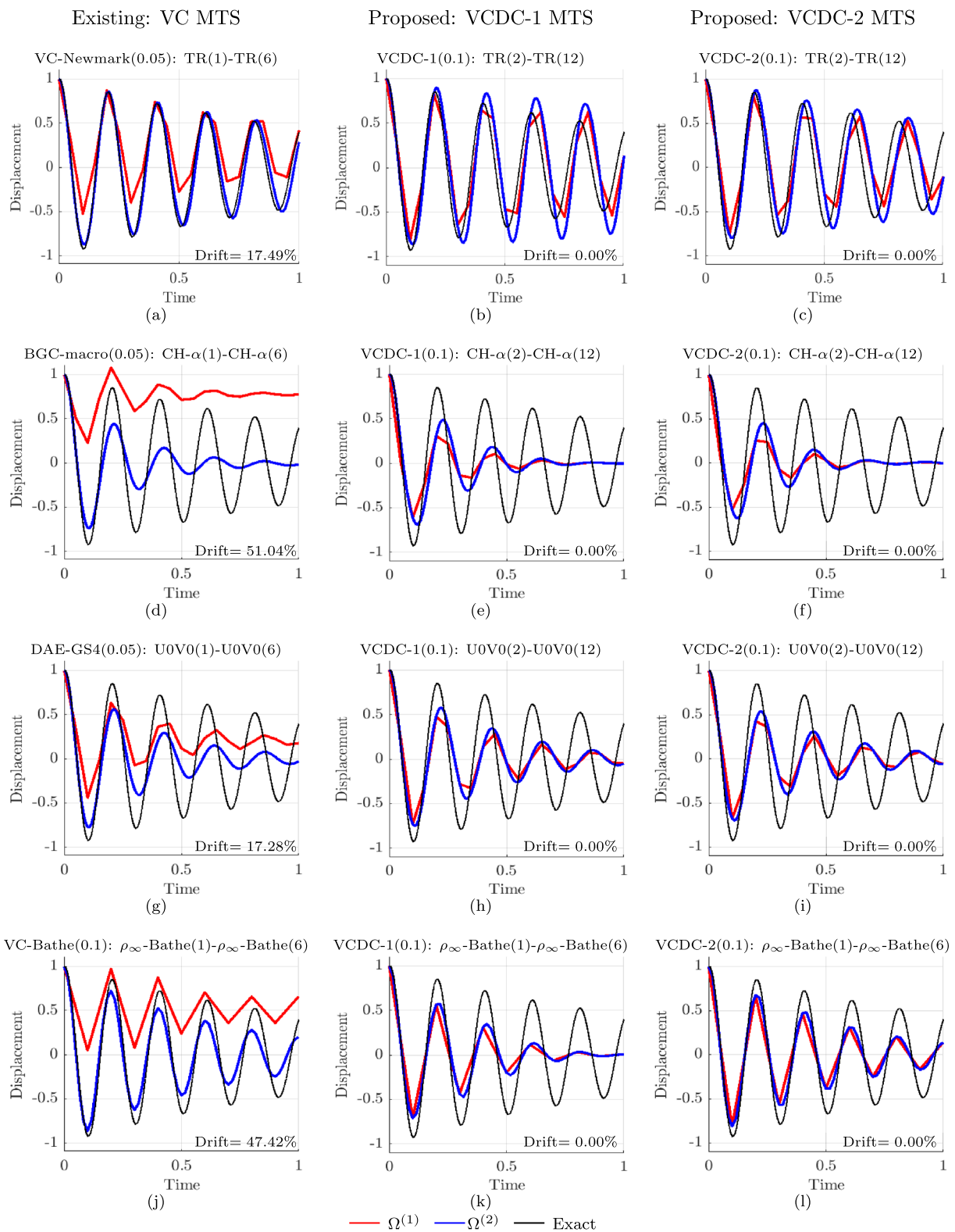
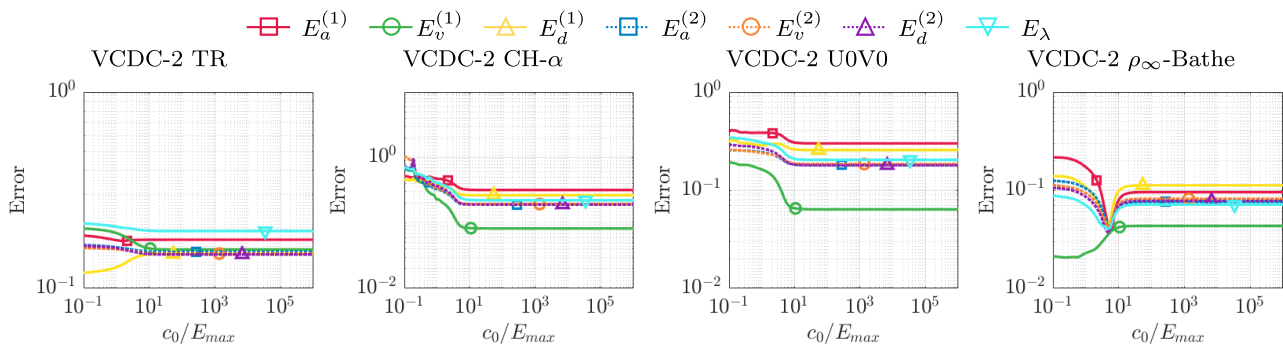


Fig. 8 Comparison of drifts in displacement for the split-SDOF problem obtained using different existing and proposed MTS methods. Different MTS methods are denoted as $\langle \text{MTS}_{\text{name}} \rangle (\Delta T)$: GRKN($m^{(1)}$)-GRKN($m^{(2)}$)

Table 6 Comparison of drifts and errors in the split-SDOF problem for existing and proposed new MTS methods

	Time steps	Drift* (%)	Error(%)			
Existing MTS methods	$(\Delta T, \Delta t^{(1)}, \Delta t^{(2)})$	(D_d, D_v, D_a)	$(E_d^{(1)}, E_d^{(2)})$	$(E_v^{(1)}, E_v^{(2)})$	$(E_a^{(1)}, E_a^{(2)})$	E_λ
VC-Newmark TR [46]	(0.05, 0.05, 0.05/6)	(17.49, 0.00, 36.25)	(9.95, 4.83)	(4.96, 5.13)	(9.93, 6.00)	10.42
BGC-macro CH- α [54]	(0.05, 0.05, 0.05/6)	(51.04, 0.00, 29.70)	(37.73, 15.54)	(14.10, 16.03)	(20.75, 16.09)	17.17
DAE-GS4 U0V0 [56]	(0.05, 0.05, 0.05/6)	(17.28, 0.00, 29.93)	(17.05, 14.10)	(12.70, 14.70)	(19.21, 14.39)	15.98
VC-Bathe [70]	(0.10, 0.10, 0.10/6)	(47.42, 0.00, 9.60)	(24.73, 6.79)	(7.06, 7.13)	(6.20, 7.08)	5.79
Proposed MTS methods	$(\Delta T, \Delta t^{(1)}, \Delta t^{(2)})$	(D_d, D_v, D_a)	$(E_d^{(1)}, E_d^{(2)})$	$(E_v^{(1)}, E_v^{(2)})$	$(E_a^{(1)}, E_a^{(2)})$	E_λ
VCDC-1 TR	(0.10, 0.05, 0.05/6)	(0.00, 0.00, 57.97)	(11.33, 13.85)	(13.50, 14.39)	(20.70, 15.42)	21.32
VCDC-1 CH- α	(0.10, 0.05, 0.05/6)	(0.00, 0.00, 16.46)	(16.16, 17.90)	(17.75, 18.34)	(22.43, 18.30)	22.08
VCDC-1 U0V0	(0.10, 0.05, 0.05/6)	(0.00, 0.00, 7.87)	(15.38, 16.25)	(14.39, 16.80)	(21.36, 16.56)	19.73
VCDC-1 ρ_∞ -Bathe	(0.10, 0.10, 0.10/6)	(0.00, 0.00, 19.05)	(18.46, 12.65)	(5.01, 13.20)	(17.16, 12.51)	12.99
VCDC-2 TR	(0.10, 0.05, 0.05/6)	(0.00, 0.00, 56.17)	(13.85, 14.94)	(14.47, 15.09)	(19.20, 15.34)	19.59
VCDC-2 CH- α	(0.10, 0.05, 0.05/6)	(0.00, 0.00, 15.76)	(17.07, 19.16)	(18.83, 19.55)	(22.58, 19.28)	22.18
VCDC-2 U0V0	(0.10, 0.05, 0.05/6)	(0.00, 0.00, 7.31)	(16.95, 18.01)	(16.00, 18.56)	(21.82, 18.08)	20.38
VCDC-2 ρ_∞ -Bathe	(0.10, 0.10, 0.10/6)	(0.00, 0.00, 16.01)	(11.22, 7.79)	(4.33, 8.17)	(9.57, 7.62)	7.13

* Existing VC MTS methods lead to drift in displacement (D_d), whereas the proposed VCDC MTS methods achieve zero drift in displacement and velocity

**Fig. 9** Errors for the split-SDOF problem for 101 different c_0/E_{max} between 10^{-1} to 10^6

methods achieve zero drift in both displacements and velocities simultaneously at every block time step ΔT .

To further evaluate the performance of existing and new MTS methods, we define the following measures of error in kinematic quantities ($E_x^{(z)}$) and Lagrange multipliers (E_λ):

$$E_x^{(z)} = \frac{\sum_{\ell=0}^{N^{(z)}} |x_\ell^{(z)} - \tilde{x}^{(z)}(t_\ell^{(z)})|}{(N^{(z)} + 1) \left(\max_{t \in [0, T]} \tilde{x}^{(z)}(t) - \min_{t \in [0, T]} \tilde{x}^{(z)}(t) \right)}, \quad (63)$$

$$E_\lambda = \frac{\sum_{k=0}^{N-1} \sum_{\beta \in \mathcal{B}} |\lambda(t_k + t_\beta) - \tilde{\lambda}(t_k + t_\beta)|}{(p^I N + 1) \left(\max_{t \in [0, T]} \tilde{\lambda}(t) - \min_{t \in [0, T]} \tilde{\lambda}(t) \right)}$$

where $N^{(z)} = m^{(z)} N$ is the total number of $\Delta t^{(z)}$ time steps over the duration of the simulation. Similarly, $p^I N$ is the total number of discrete time instances on the timeline

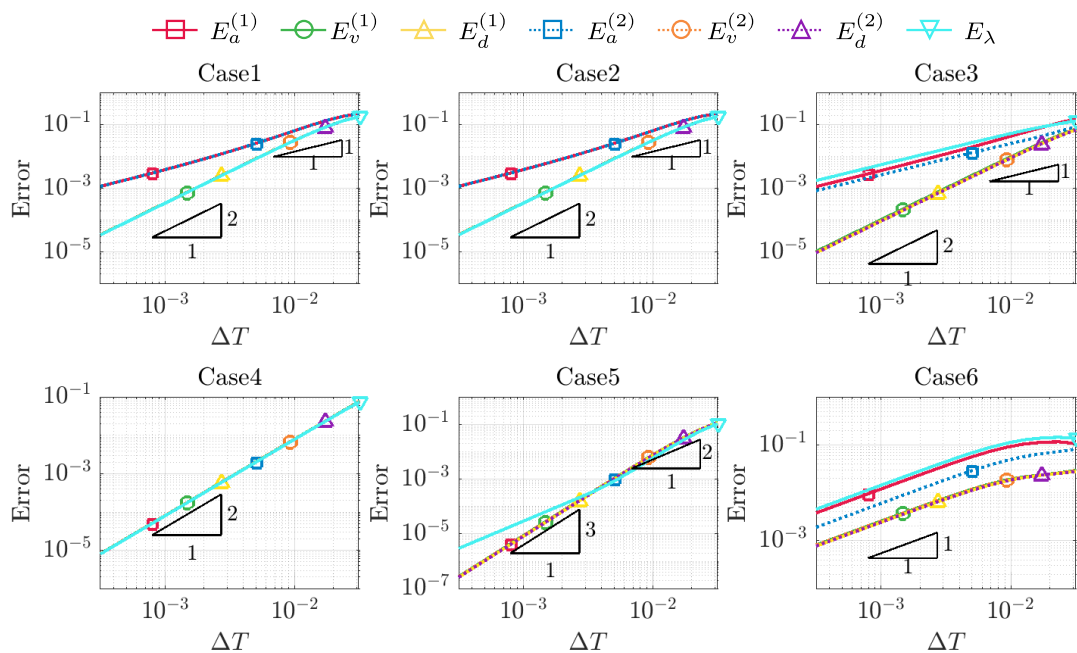
for the interface Γ^I as shown in Fig. 4. Drifts and errors associated with existing and new MTS methods for the split-SDOF problem are summarized in Table 6. Note that the proposed VCDC MTS methods maintain similar levels of error as the existing VC MTS methods while achieving zero drift in displacements (D_d) and velocities (D_v).

For the proposed MTS methods listed in Table 6, SAV-based stabilization is used where the maximum energy is estimated as $E_{max} = 1/2 \sum_{z=1}^2 \left(M^{(z)} v_0^2 + M^{(z)} \omega_0^{(z)^2} d_0^2 \right)$ and c_0 is selected as $10^4 E_{max} = 2 \times 10^7$ based on empirical observations presented in Fig. 9 which shows errors for various values of c_0/E_{max} . Note that errors depend upon c_0 , but stabilize for values of $c_0/E_{max} \geq 10^2$. For other values of ΔT and m for different MTS methods, a conservative value of $c_0 = 10^4 E_{max}$ is found to be sufficient. In general, however, the value of c_0 is problem dependent and requires the user to conduct some exploratory studies, similar to those presented in Fig. 9 to determine an appropriate value.

Table 7 Convergence rates of proposed VCDC MTS methods for different GRKN schemes for the split-SDOF problem

Case	Underlying Time integration		Convergence rate $(E_a^{(1)}, E_v^{(1)}, E_d^{(1)}, E_a^{(2)}, E_v^{(2)}, E_d^{(2)}, E_\lambda^I)$			
	$\Omega^{(1)}$	$\Omega^{(2)}$	$m^{(2)}/m^{(1)} = 1$		$2 \leq m^{(2)}/m^{(1)} \leq 10$	
			VCDC-1	VCDC-2	VCDC-1	VCDC-2
1	CH- α^a	CH- α^a	(1,2,2,1,2,2,2)	(1,2,2,1,2,2,2)	(1,2,2,1,2,2,1)	(1,1,1,1,1,1,1)
2	CH- α^a	WBZ- α^a	(1,2,2,1,2,2,2)	(1,2,2,1,2,2,2)	(1,2,2,1,2,2,1)	(1,1,1,1,1,1,1)
3	CH- α^a	Central difference (CD)	(1,2,2,1,2,2,1)	(1,2,2,1,2,2,1)	(1,2,2,1,2,2,1)	(1,1,1,1,1,1,1)
4	ρ_∞ -Bathe ^b	ρ_∞ -Bathe ^b	(2,2,2,2,2,2,2)	(2,2,2,2,2,2,2)	(1,1,1,1,1,1,1)	(1,1,1,1,1,1,1)
5	ρ_∞ -Bathe ^c	ρ_∞ -Bathe ^c	(3,3,3,3,3,3,2)	(3,3,3,3,3,3,3)	(2,2,2,2,2,2,2)	(2,3,3,2,3,3,2)
6	ρ_∞ -Bathe ^b	Noh-Bathe ^d	(1,1,1,1,1,1,1)	(1,1,1,1,1,1,1)	(1,1,1,1,1,1,1)	(1,1,1,1,1,1,1)

^a $\rho_\infty = 0$; ^b $\rho_\infty = 0.65$ and $\gamma_0 = \frac{2 - \sqrt{2 + 2\rho_\infty}}{1 - \rho_\infty}$; ^c $\rho_\infty = 1 - \sqrt{3}$ and $\gamma_p = \frac{\rho_\infty + 2 - \sqrt{\rho_\infty^2 - 2\rho_\infty - 2}}{3(\rho_\infty + 1)}$; and ^d $q = 0.54$

**Fig. 10** Errors using the VCDC-1 method for the split-SDOF problem with $m^{(2)}/m^{(1)} = 1$

Finally, we carry out a series of convergence tests for the split-SDOF problem using six different proposed MTS methods (listed in Table 7) with different underlying time integration schemes. We use time-step ratio $m^{(1)} = 1$ (for composite schemes) and $m^{(1)} = 2$ (for single-stage schemes) while varying time-step ratio $m^{(2)}$ such that $m^{(2)}/m^{(1)} = 1, 2, \dots, 10$. Figures 10, 11 and 12 show errors for the proposed MTS methods with $m^{(2)}/m^{(1)} = 1, 3, 6$, and 10. The rate of convergence of these MTS methods is also summarized in Table 7. Note that VCDC-1 preserves the convergence rate of the underlying single-stage schemes that are second-order accurate (Cases 1, 2, and 3). In contrast, VCDC-2 only preserves the convergence rates when the underlying GRKN scheme is a third-order

accurate composite scheme (Case 5). Even though the convergence rates for some of the MTS methods presented here do not match those of the underlying UTS schemes (Cases 4 and 6), the proposed MTS framework makes it possible to design new MTS methods (by defining alternative constraint conditions) that may be able to achieve optimal convergence rates - an exercise that we leave for the future.

4.2 2D scalar wave propagation with two different media

In this section, the problem of a two-dimensional (2D) scalar wave propagation in a pre-stressed membrane (see Fig. 13) is solved using existing MTS methods and new

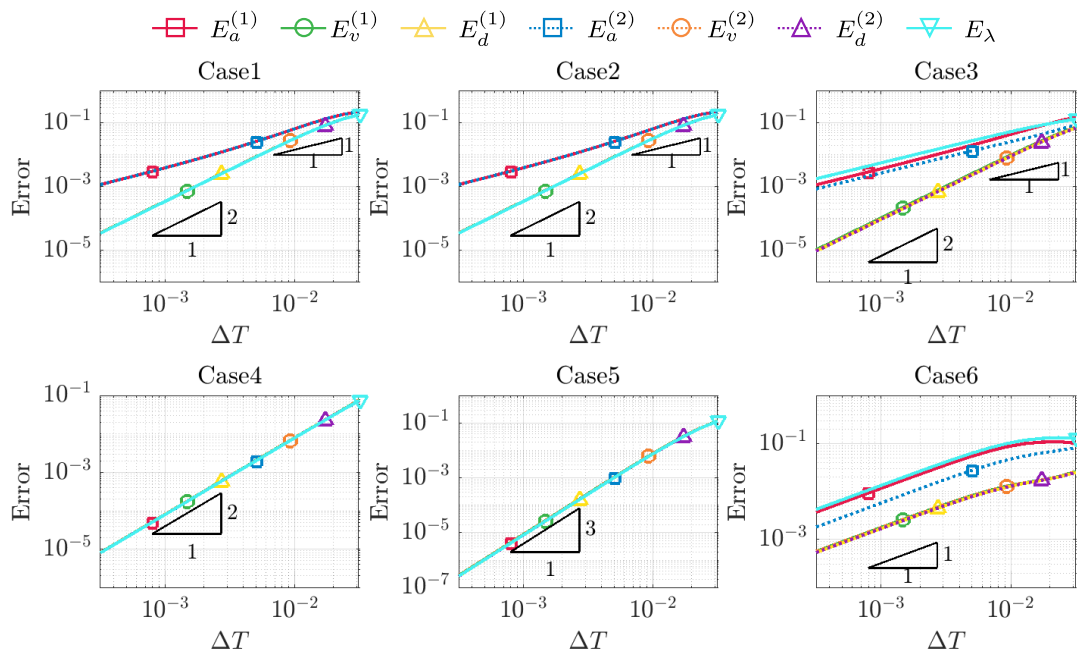


Fig. 11 Errors using the VCDC-2 method for the split-SDOF problem with $m^{(2)}/m^{(1)} = 1$

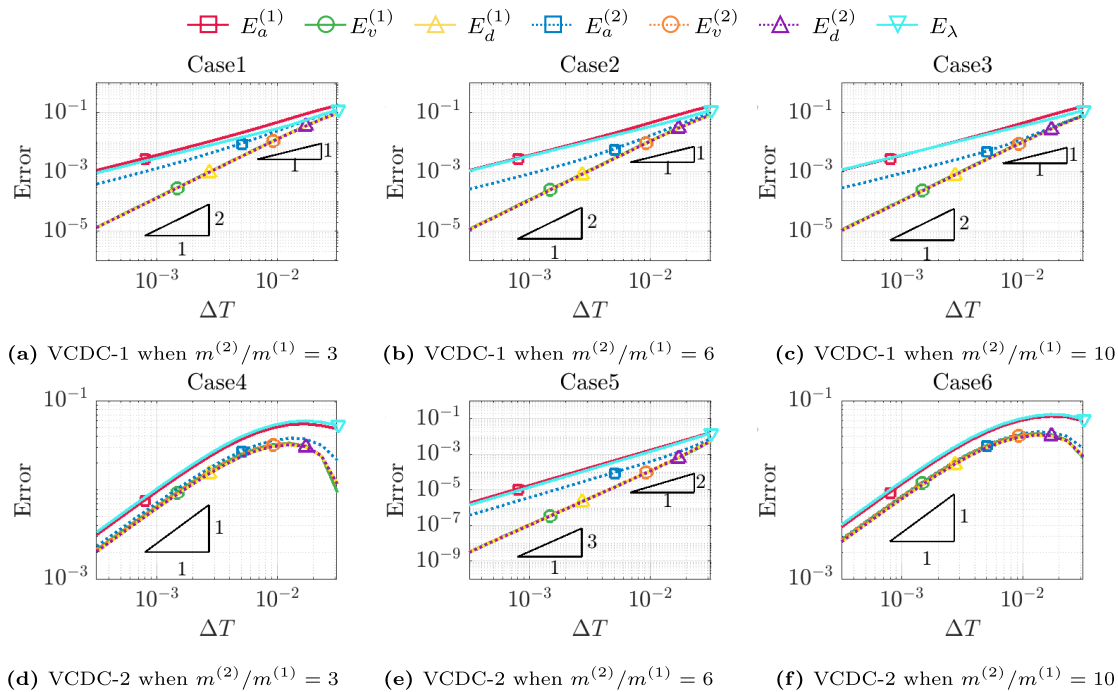


Fig. 12 Errors using the VCDC methods for the split-SDOF problem with $m^{(2)}/m^{(1)} = 3, 6$, and 10

MTS methods designed using the proposed MTS framework. This problem is often used to evaluate the characteristics of time integration schemes for wave propagation [21, 77, 78]. In this problem, initial displacement and velocity are set to zero for the entire problem domain and the load at the center of the membrane is given as

$$f(0, 0, t) = 4 \left(1 - (2t - 1)^2 \right) H(1 - t), \quad t > 0 \quad (64)$$

where H is the Heaviside step function.

Since the membrane is symmetric about the x and y axes, only a quarter of the domain $[0, 20]$ by $[0, 20]$ is discretized using a structured mesh of $N^{el} \times N^{el}$ 4-noded quadrilateral elements with equally spaced nodes. We consider 5 levels of

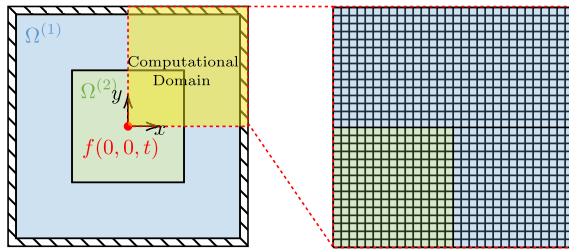


Fig. 13 Pre-stressed membrane with two different media. Wave speed: $u_w^{(1)} = 1$, $u_w^{(2)} = 4$; Loading: $f(0, 0, t) = 4(1 - (2t - 1)^2)H(1 - t)$; Initial conditions: $\mathbf{u}(0) = \dot{\mathbf{u}}(0) = \mathbf{0}$

mesh refinement with $N^{el} = 30, 50, 100, 200$, and 300 . The computational domain is divided into two subdomains $\Omega^{(1)}$ and $\Omega^{(2)}$ integrated with time steps $\Delta t^{(1)}$ and $\Delta t^{(2)}$, respectively. These time steps in the two subdomains are selected based on the wave speeds, the optimal CFL (Courant-Friedrichs-Lewy) numbers associated with the underlying time integration, and the characteristic lengths of elements: $\Delta t^{(z)} = \text{CFL} \times h^{(z)} / u_w^{(z)}$, where $h^{(z)}$ and $u_w^{(z)}$ denote the characteristic length and wave speed, respectively, in subdomain $\Omega^{(z)}$. Characteristic length is chosen to be the length of a side of the square elements. Two time integration schemes are used, the trapezoidal rule (TR) with $\text{CFL} = 0.65$, and the ρ_∞ -Bathe scheme [20] ($\rho_\infty = 0.65$ and $\gamma = \gamma_0$) with $\text{CFL} = 1.25$, both with their optimal CFL numbers [31]. The time

duration considered is $T = 8$, for which the wave does not reach the outer boundary of the problem domain.

The problem is solved using three approaches: (1) Uniform time-step (UTS) method with the small time step $\Delta t^{(2)}$ for the entire problem domain, (2) Existing VC MTS methods with time-step ratios $m^{(1)} = 1$ and $m^{(2)} = 4$, and (3) new VCDC MTS methods with quadratic interpolation (VCDC-2) also with time-step ratios $m^{(1)} = 1$ (for composite schemes) or 2 (for single-stage schemes) and $m^{(2)} / m^{(1)} = 4$. For time-step parity, the block time step ΔT for the VCDC-2 MTS method using the TR scheme is taken to be twice that of the existing VC MTS method for the TR scheme. Note that, with the UTS approach, it is not possible to use optimal CFL numbers with a uniform mesh because of the difference in wave speeds between $\Omega^{(1)}$ and $\Omega^{(2)}$. On the other hand, MTS methods enable the use of optimal CFL numbers in each subdomain, thereby minimizing errors due to numerical dispersion.

For SAV stabilization of the VCDC MTS methods, we select $c_0 = 10^4 \times E_{max}$. We estimate E_{max} as $\frac{1}{2} \mathbf{d}_{max}^T \mathbf{K} \mathbf{d}_{max}$, where maximum displacement $\mathbf{d}_{max}$ is assumed to be twice the static response (i.e. $\mathbf{d}_{max} \approx 2\mathbf{K}^{-1} \max(\mathbf{f})$).

To evaluate accuracy of the solution, we define the following L^2 norm of error:

$$L_2(\mathbf{x}) = \begin{cases} \sqrt{\frac{\sum_{n=0}^N (\mathbf{x}_{4n} - \tilde{\mathbf{x}}(t_{4n}))^T \mathbf{M} (\mathbf{x}_{4n} - \tilde{\mathbf{x}}(t_{4n}))}{\sum_{n=0}^N \tilde{\mathbf{x}}(t_{4n})^T \mathbf{M} \tilde{\mathbf{x}}(t_{4n})}} & \text{for UTS methods} \\ \sqrt{\frac{\sum_{n=0}^N \sum_{z=1}^S (\mathbf{x}_{mn}^{(z)} - \tilde{\mathbf{x}}(t_{mn}^{(z)}))^T \mathbf{M}^{(z)} (\mathbf{x}_{mn}^{(z)} - \tilde{\mathbf{x}}(t_{mn}^{(z)}))}{\sum_{n=0}^N \sum_{z=1}^S \tilde{\mathbf{x}}(t_{mn}^{(z)})^T \mathbf{M}^{(z)} \tilde{\mathbf{x}}(t_{mn}^{(z)})}} & \text{for MTS methods} \end{cases} \quad (65)$$

Table 8 Computational gains, L^2 errors, and drifts for the scalar wave propagation problem with element size $h = 0.2$

	Computational gain		Error		Drift (%)
	Actual	Estimate	$L_2(\mathbf{d})$	$L_2(\mathbf{v})$	D_d
Baseline UTS schemes					
TR(0.0325) [5]	1.00	1.00	0.09	0.36	0.00
ρ_∞ -Bathe(0.0625) [20]	1.00	1.00	0.09	0.35	0.00
Existing MTS methods					
VC-Newmark(0.13): TR(1)-TR(4) [46]	2.25	1.94	0.07	0.26	3.05
VC-Bathe(0.25):	3.16	2.41	0.19	0.28	3.28
ρ_∞ -Bathe(1)- ρ_∞ -Bathe(4) [70]					
Proposed MTS methods					
VCDC-2(0.26): TR(2)-TR(8)	2.08	1.97	0.03	0.24	0.00
VCDC-2(0.25):	2.35	1.98	0.03	0.18	0.00
ρ_∞ -Bathe(1)- ρ_∞ -Bathe(4)					

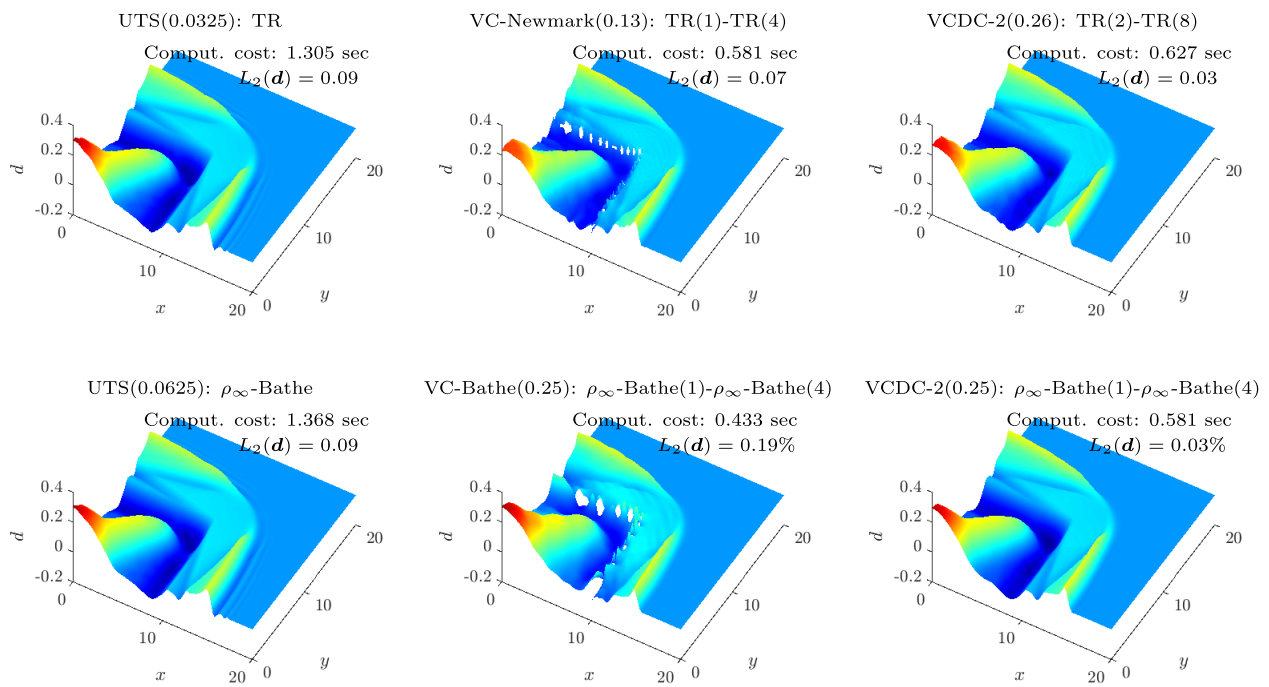


Fig. 14 Snapshots of displacement at time $t = 8$ for the scalar wave propagation problem when $h = 0.2$

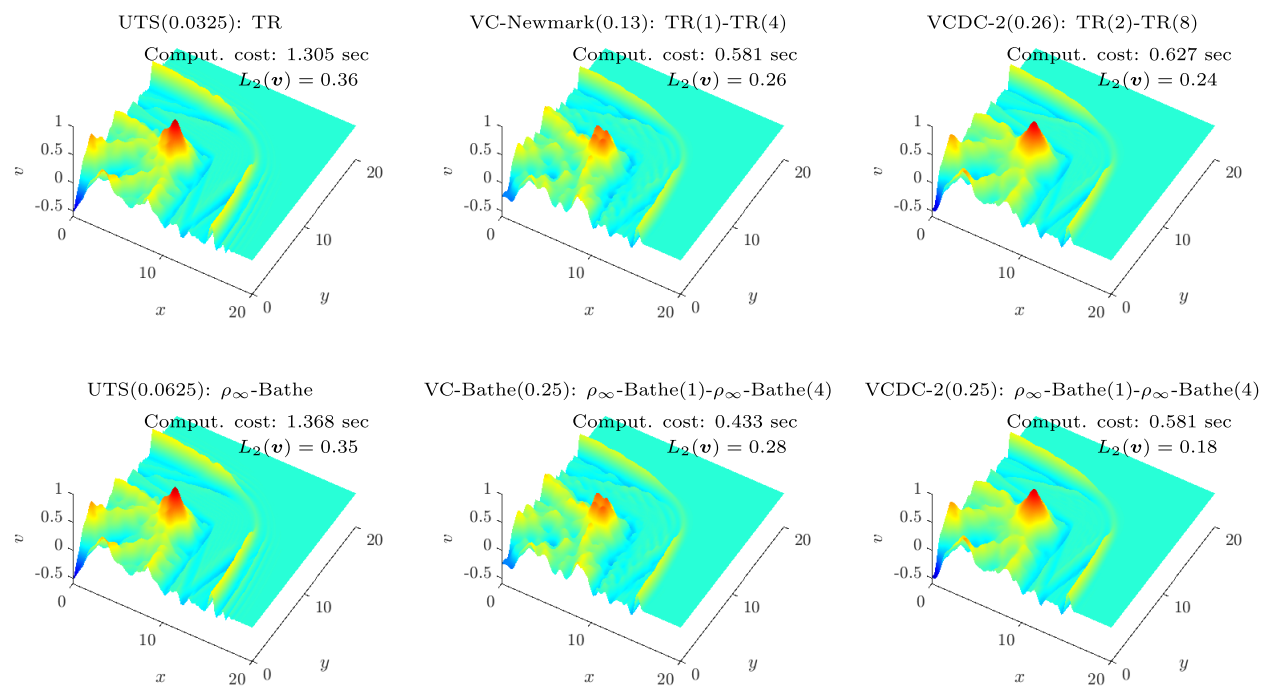


Fig. 15 Snapshots of velocity at time $t = 8$ for the scalar wave propagation problem when $h = 0.2$

where $\mathbf{x}$ is a quantity of interest (e.g. displacement $\mathbf{d}$ or velocity $\mathbf{v}$), and $\tilde{\mathbf{x}}(t)$ is its reference solution. Note that the errors norms for the UTS and MTS methods have been defined to measure differences at block-time-step intervals ΔT . Reference solutions for this problem are obtained using existing and proposed MTS methods for the Bathe scheme with a very fine

mesh ($h = 0.02$) and correspondingly small time steps chosen according to the CFL numbers of the subdomains. L^2 norms of the differences between these reference solutions are found to be of the order of 10^{-4} in displacement and 10^{-2} in velocity which are much smaller than the L^2 norms of errors reported in Table 8, justifying their use as reference solutions.

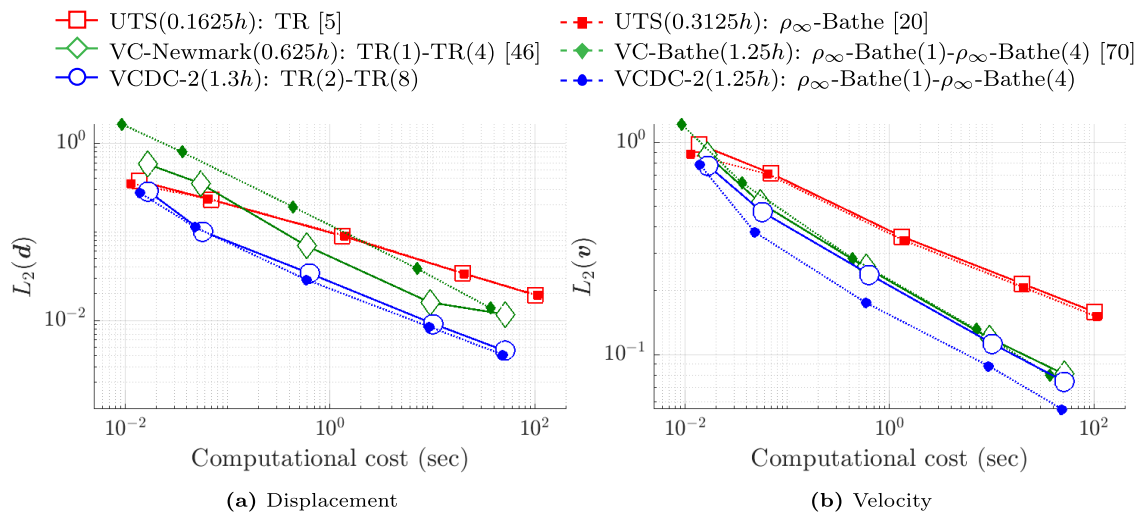


Fig. 16 L^2 norm errors in displacement and velocity for the scalar wave propagation problem. The 5 points on each curve correspond to different mesh sizes $h = 2/3, 0.4, 0.2, 0.1$, and $2/30$ with increasing computational costs from left to right

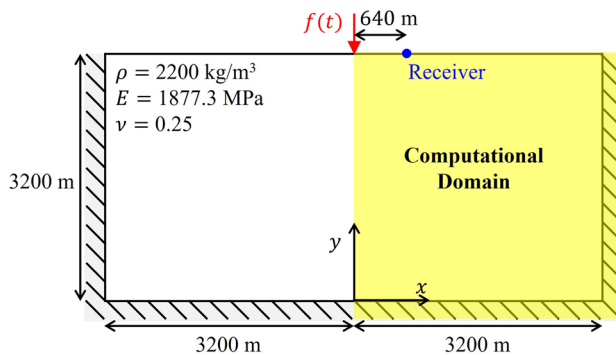


Fig. 17 2D Lamb's problem in the plane strain conditions. A receiver is located at $(x, y) = (640, 3200)$ m

Figures 14 and 15 show solutions for the displacement and velocity, respectively, at the end of the simulation time $T = 8$ using the UTS approach, existing MTS methods, and proposed MTS methods for TR and ρ_∞ -Bathe schemes. Note that existing MTS methods lead to drifts in displacements that show up as gaps and overlaps, whereas the proposed VDCDC-2 MTS methods are able to achieve zero drift. Further, since the CFL numbers are not optimal for the UTS approach, it leads to spurious oscillations ahead of the wave front. On the other hand, solutions from existing and proposed MTS methods do not suffer from these spurious oscillations and lead to accurate results because the time steps used in both subdomains are optimal for wave propagation.

Table 8 lists computational gains, errors, and drifts of MTS methods for element size $h = 0.2$. We note that actual computational gains listed in Table 8 match well with the estimated gains obtained using the computational cost estimates from Sect. 3.6. Additionally, note that the proposed VDCDC-2 MTS methods run about twice faster than the underlying UTS

schemes, while maintaining similar levels of accuracy and achieving zero drift in displacements.

Figure 16 compares the computational cost and errors in displacement and velocity for different UTS and MTS methods for five levels of spatial discretization. Note that the proposed VDCDC-2 MTS methods are able to achieve lower errors compared to the underlying UTS and existing VC MTS approaches for similar computational costs.

4.3 2D Lamb's problem with two levels of spatial discretization

We solve the Lamb's problem shown in Fig. 17 where multiple waves are propagating in an elastic medium under plane strain conditions (see [18, 77–81]). In this problem, we choose the following material properties: mass density $\rho = 2200 \text{ kg/m}^3$, Poisson's ratio $\nu = 0.25$, and elastic modulus $E = 18,773 \text{ MPa}$. The P- and S-wave velocities are $u_p = 3200 \text{ m/s}$ and $u_s = 1847.5 \text{ m/s}$, respectively. The applied force is in the form of a Ricker wavelet: $f(t) = 2 [1 - 2\pi^2 f_c^2 (t - t_0)^2] e^{-\pi^2 f_c^2 (t - t_0)^2}$ MN/m, where the central frequency $f_c = 10 \text{ Hz}$ and the time shift $t_0 = 0.1 \text{ sec}$. As the load is concentrated at the origin, we consider $\Omega = [0, 3200] \text{ m} \times [0, 3200] \text{ m}$ as the computational domain. Initial displacement and velocity in the domain are zero and we measure displacement at the receiver located at $(x, y) = (640, 3200) \text{ m}$. Note that, for the time duration considered $T = 1 \text{ sec}$, the wave does not reach the outer boundary of the problem domain.

We divide the problem domain into two subdomains: $\Omega^{(2)} = [0, 740] \text{ m} \times [3100, 3200] \text{ m}$ and $\Omega^{(1)} = \Omega \setminus \Omega^{(2)}$, as shown in Fig. 18. For spatial discretization, a structured mesh is used with 4-noded quadrilateral (Q4) finite elements. In

Fig. 18 Spatial mesh discretization for the 2D Lamb's problem with two subdomains. In subdomains $\Omega^{(1)}$ and $\Omega^{(2)}$, the characteristic lengths are $h^{(1)} = 20$ m and $h^{(2)} = 2$ m, respectively

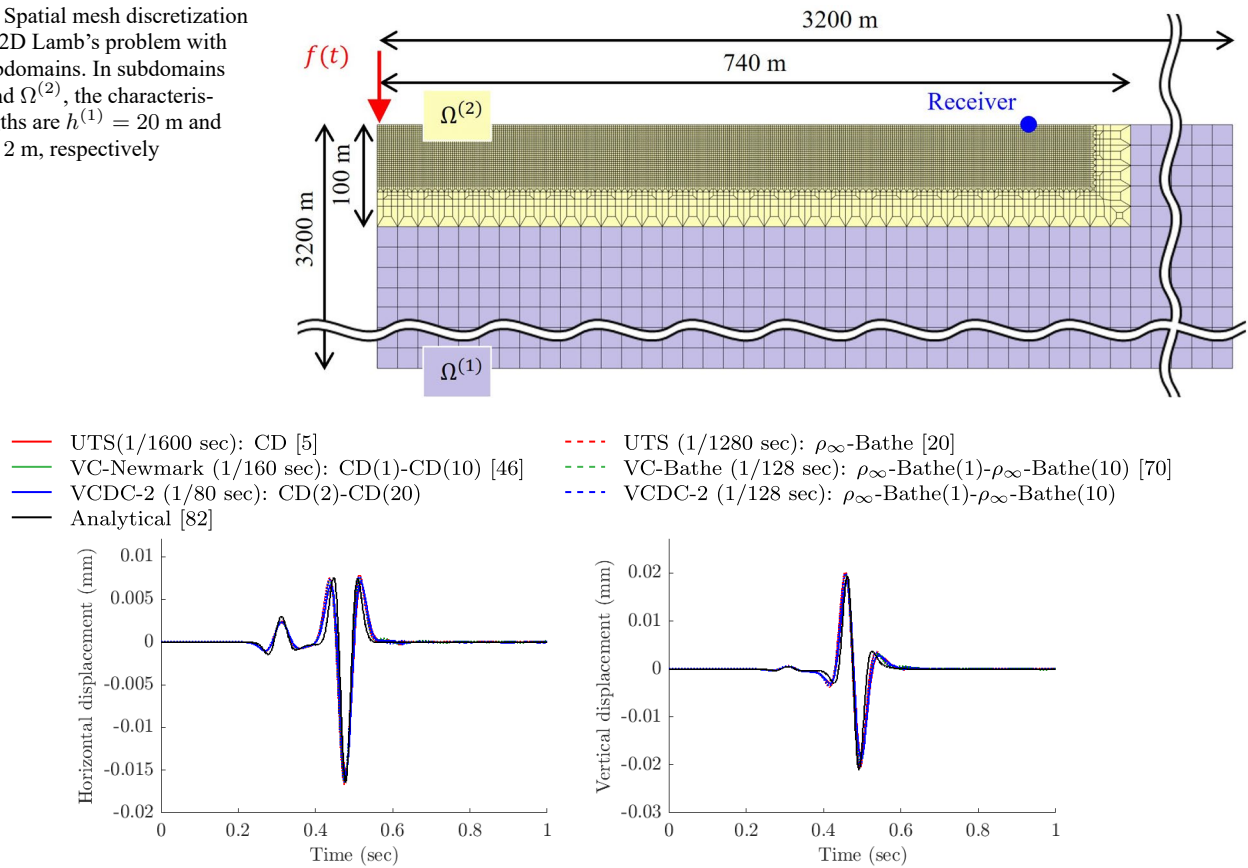


Fig. 19 Time history of horizontal and vertical displacements at $(x, y) = (640, 3200)$ m for the 2D Lamb's problem

subdomain $\Omega^{(1)}$, we use 25,415 square Q4 elements with $h^{(1)} = 20$ m. Subdomain $\Omega^{(2)}$ is discretized with 1,056 trapezoid Q4 elements and 10,500 square Q4 elements with $h^{(2)} = 2$ m. Similar to Sect. 4.2, we select time steps based on P-wave velocity, the optimal CFL number, and the length of the side of square Q4 elements: $\Delta t^{(z)} = \text{CFL} \times h^{(z)} / u_p$ (i.e. $\Delta t^{(1)} = \text{CFL}/160$ sec and $\Delta t^{(2)} = \text{CFL}/1600$ s).

The problem is solved using two different time integration schemes, the CD scheme [5] (with $\gamma = 0.5$, $\beta = 0$, and $\text{CFL} = 1$) and the ρ_∞ -Bathe scheme [20] (with $\rho_\infty = 0.65$, $\gamma = \gamma_0$, and $\text{CFL} = 1.25$). For time discretization, three different approaches are adopted: (i) the UTS method with $\Delta t^{(2)}$ for the entire domain, (ii) the VC MTS method with time-step ratios $m^{(1)} = 1$ and $m^{(2)} = 10$, and (iii) the proposed VCDC MTS method

with $m^{(1)} = 1$ (for composite schemes) or $m^{(1)} = 2$ (for single-stage schemes) and $m^{(2)}/m^{(1)} = 10$. Note that for MTS methods, we choose the time-step ratio $m^{(2)}/m^{(1)} = 10$ because of the spatial discretization with $h^{(1)}/h^{(2)} = 10$. For SAV stabilization of the proposed VCDC MTS methods, the maximum energy is approximated as $E_{\max} \approx \frac{1}{2} \mathbf{d}_{\max}^T \mathbf{K} \mathbf{d}_{\max} \approx 151.3$ J/m during the simulation, where $\mathbf{d}_{\max} \approx 2[\mathbf{K}]^{-1} \max(\mathbf{f})$. As in the previous examples, we choose the SAV parameters $c_0 = 10^4 E_{\max} = 1513$ kJ/m.

To evaluate accuracy of the solution at the receiver, we define the following measure of local cumulative error in time:

$$\epsilon(x) = \begin{cases} \frac{\sum_{n=0}^N |x_n - \tilde{x}(t_n)|}{(N+1) \left(\max_{t \in [0, T]} \tilde{x}(t) - \min_{t \in [0, T]} \tilde{x}(t) \right)} & \text{for UTS approach} \\ \frac{\sum_{\ell=0}^{N^{(z)}} |x_\ell^{(z)} - \tilde{x}(t_\ell^{(z)})|}{(N^{(z)}+1) \left(\max_{t \in [0, T]} \tilde{x}(t) - \min_{t \in [0, T]} \tilde{x}(t) \right)} & \text{for MTS methods} \end{cases} \quad (66)$$

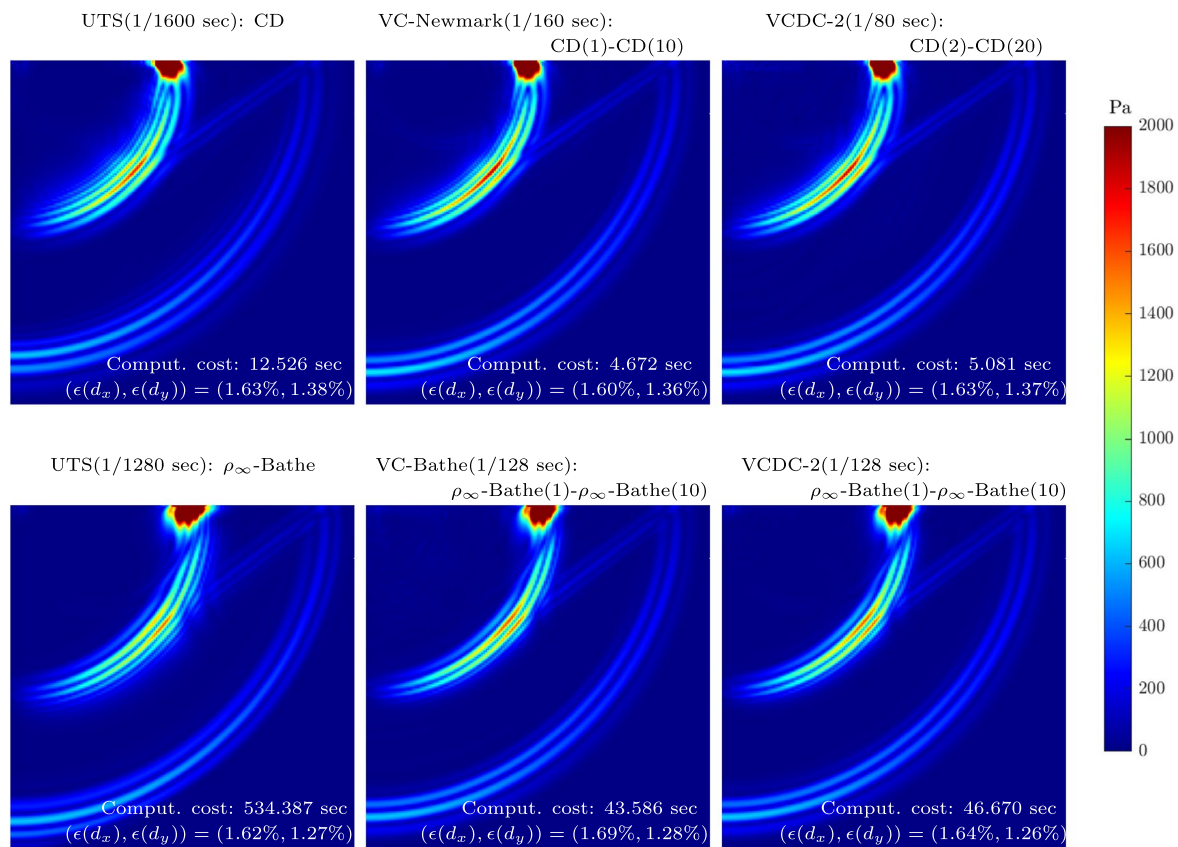


Fig. 20 Snapshots of von Mises stress at time $t = 1$ sec for the 2D Lamb's problem

Table 9 Computational gains, local cumulative errors at $(x, y) = (640, 3200)$ m, and relative drifts at $(x, y) = (740, 3200)$ m for the 2D Lamb's problem

	Comp. gain	Error (%)		Drift (%)	
		$\epsilon(d_x)$	$\epsilon(d_y)$	D_{d_x}	D_{d_y}
Baseline UTS schemes	Actual				
CD(1/1600 s) [5]	1.00	1.63	1.38	0.00	0.00
ρ_∞ -Bathe(1/1280 s) [20]	1.00	1.62	1.27	0.00	0.00
Existing MTS methods	Actual	$\epsilon(d_x)$	$\epsilon(d_y)$	D_{d_x}	D_{d_y}
VC-Newmark (1/160 s): CD(1)-CD(10) [46]	2.68	1.60	1.36	2.36	1.96
VC-Bathe(1/128 s): ρ_∞ -Bathe(1)- ρ_∞ -Bathe (10) [70]	12.26	1.69	1.28	2.54	0.51
Proposed MTS methods	Actual	$\epsilon(d_x)$	$\epsilon(d_y)$	D_{d_x}	D_{d_y}
VCDC-2(1/80 s): CD(2)-CD(20)	2.47	1.63	1.37	0.00	0.00
VCDC-2(1/128 s): ρ_∞ -Bathe(1)- ρ_∞ -Bathe(10)	11.45	1.64	1.26	0.00	0.00

where N is the number of time-steps, x may be the horizontal or vertical displacement at the receiver d_x or d_y , and $\tilde{x}(t)$ is the analytical solution given in [82]. Note that for the MTS methods, $N^{(z)}$ is taken as $N^{(2)}$ since the receiver is always in subdomain $\Omega^{(2)}$.

Figure 19 compares numerical solutions at the receiver for the two time integration schemes and three time discretizations. Figure 20 shows von Mises stress distributions at time $t \approx 1$ sec, computational costs, and local cumulative errors at the receiver. Similar to the previous example, the UTS approach leaves spurious oscillations behind the wave front for the CD scheme and ahead of the wave front for the ρ_∞ -Bathe scheme. This is because, with the UTS method, the CFL number cannot be optimal for both subdomains simultaneously, specifically for $\Omega^{(1)}$ in this case. On the other hand, the existing VC MTS and the proposed VCDC-2 MTS methods yield accurate solutions without visible numerical oscillations. Note that the existing VC MTS methods lead to drifts in displacement whereas the VCDC MTS methods are able to achieve similar accuracy with zero drifts in displacement. Lastly, as shown in Table 9, the MTS methods are approximately 2.5 and 12 times faster than the UTS approach (i.e. computational gain ≈ 2.5 for the explicit CD scheme and ≈ 12 for the implicit ρ_∞ -Bathe scheme).

4.4 Impact of a sandwich plate

Having considered implicit-implicit MTS methods and explicit-explicit MTS methods in the previous two examples,

Fig. 21 Impact of a sandwich plate. Material properties of stiff plates: mass density $\rho = 27 \times 10^{-10}$ ton/mm³, Poisson's ratio $\nu = 0.3$, and elastic modulus $E = 70,000$ MPa. Material properties of the soft core: mass density $\rho = 9 \times 10^{-10}$ ton/mm³, Poisson's ratio $\nu = 0.3$, and elastic modulus $E = 7,780$ MPa. A smooth step function with $t_0 = 10^{-5}$ sec and $p_0 = 1$ MPa is used to mimic the impact load $p(t)$

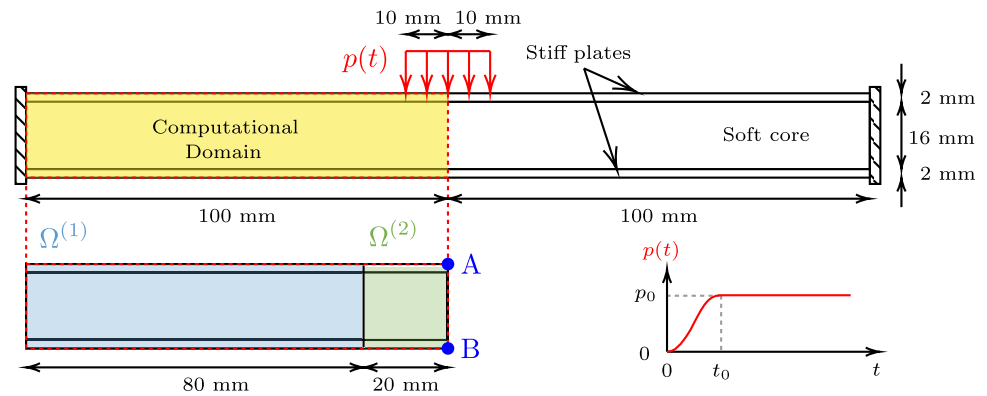


Table 10 Computational gains, L^2 norm errors, and drifts for the impact problem

	Comput. gain	Error		Drift (%)
Baseline UTS schemes	Actual	$L_2(d)$	$L_2(v)$	D_d
CD(10^{-7} s) [5]	1.00	7.5×10^{-4}	6.6×10^{-3}	0.0000
Noh-Bathe(2×10^{-7} s) [18]	1.00	7.6×10^{-4}	6.7×10^{-3}	0.0000
Existing MTS methods	Actual	$L_2(d)$	$L_2(v)$	D_d
VC-Newmark(5×10^{-7} s): TR(1)-CD(5) [46]	1.81	3.0×10^{-4}	3.2×10^{-3}	0.0052
Proposed MTS methods	Actual	$L_2(d)$	$L_2(v)$	D_d
VC-GRKN(10^{-6} s): ρ_∞ -Bathe(1)-Noh-Bathe(5)	2.21	3.4×10^{-4}	3.8×10^{-3}	0.0168
VCDC-2(10^{-6} s): TR(2)-CD(10)	1.68	3.1×10^{-4}	3.2×10^{-3}	0.0000
VCDC-2(10^{-6} s): ρ_∞ -Bathe(1)-Noh-Bathe(5)	1.92	3.3×10^{-4}	3.7×10^{-3}	0.0000

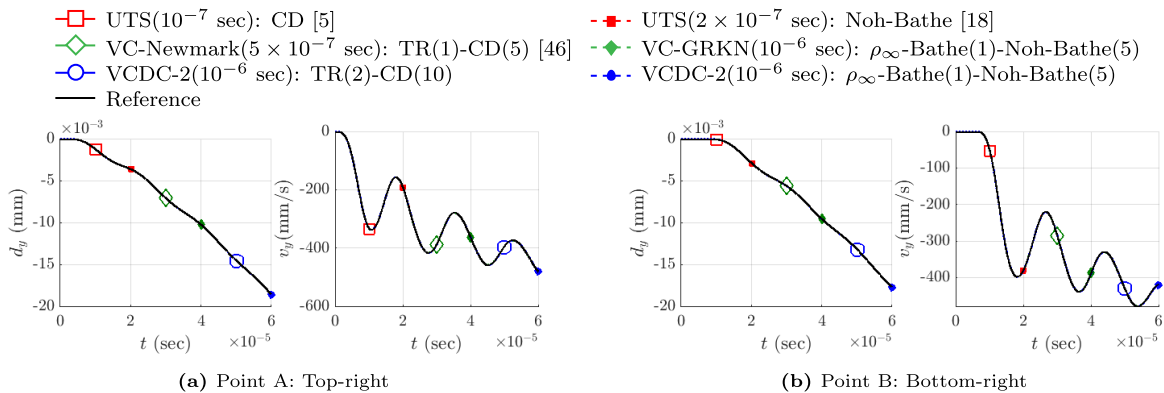


Fig. 22 Time history of vertical displacements and velocities at Points A and B of the sandwich plate under impact

we turn our attention to implicit-explicit MTS methods in this example. As shown in Fig. 21, we consider a sandwich plate under plane stress, fixed at both ends and subjected to an impact load [47]. Initial displacement and velocity are taken to be zero: $d_0 = v_0 = 0$ and damping is neglected. The impact is simulated with an external pressure ramped up from 0 to $p_0 = 1$ MPa in $t_0 = 10^{-5}$ sec using a smooth step function: $p(t) = p_0(6(t/t_0)^5 - 15(t/t_0)^4 + 10(t/t_0)^3)$ applied at the center of the top edge. Due to symmetry, we model only the left half of the problem domain. The problem domain is decomposed into two subdomains: $\Omega^{(1)}$ away from impact zone and $\Omega^{(2)}$ near the impact zone. For spatial discretization, we use a regular mesh

of Q4 elements with equally spaced nodes ($h^{(1)} = h^{(2)} = 1$ mm). Reference solutions for this problem are obtained using the UTS TR and UTS ρ_∞ -Bathe schemes with $\Delta t = 10^{-8}$ sec. We note that the L^2 norm of the difference between these two reference solutions obtained using Eq. (65) is of the order of 10^{-7} in displacement and 10^{-6} in velocity. In comparison, errors for the methods under consideration reported in Table 10 are of the order of 10^{-4} and 10^{-3} in displacement and velocity, respectively, thereby justifying the use of these reference solutions to compute errors.

The problem is solved using three different time discretizations: explicit UTS methods, implicit-explicit VC MTS methods,

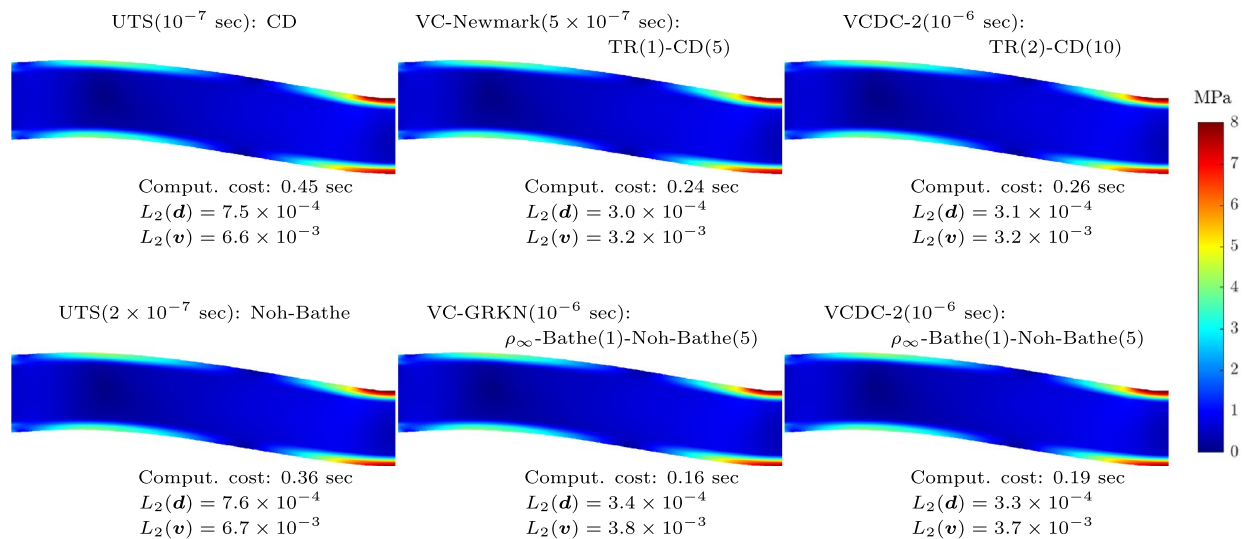


Fig. 23 Snapshots of von Mises stress and deformation (deformation scaling factor is 500) at time $t = 6 \times 10^{-5}$ sec for the impact problem

and implicit-explicit VDC-2 MTS methods. Subdomain $\Omega^{(2)}$ is solved with explicit time integrations (using the CD scheme [5] and the Noh-Bathe scheme [18] with algorithmic parameter $q = 0.54$) with a small $\Delta t^{(2)}$ to preserve accuracy and $\Omega^{(1)}$ is treated with implicit time integration (using the TR scheme [5] and the ρ_∞ -Bathe [20] with $\rho_\infty = 0.65$ and $\gamma = \gamma_0$) with $\Delta t^{(1)}/\Delta t^{(2)} = 5$ for lower computational costs. Critical time steps for the CD and Noh-Bathe schemes are 1.9092×10^{-7} sec and 3.5703×10^{-7} sec, respectively. Thus, for the UTS methods, we select a time step $\Delta t = 10^{-7}$ sec for the CD scheme and $\Delta t = 2 \times 10^{-7}$ sec for the Noh-Bathe scheme. Similarly, for the MTS methods, time step $\Delta t^{(2)} = 10^{-7}$ sec is chosen for the CD scheme and $\Delta t^{(2)} = 2 \times 10^{-7}$ sec is chosen for the Noh-Bathe scheme. Note that among the four implicit-explicit MTS methods under consideration here, only the VC-Newmark method for coupling the CD and TR schemes is an existing method in the literature. The other three methods, namely the VC-GRKN method for coupling ρ_∞ -Bathe and Noh-Bathe schemes, the VDC-2 method for coupling CD and TR schemes, and the VDC-2 method for coupling ρ_∞ -Bathe and Noh-Bathe schemes are all new MTS methods developed using the proposed framework. For SAV stabilization, we estimate the maximum energy as $E_{max} \approx \frac{1}{2} \mathbf{d}_{max}^T \mathbf{K} \mathbf{d}_{max} \approx 0.9590$ J/m, where $\mathbf{d}_{max} \approx 2[\mathbf{K}]^{-1} \max(\mathbf{f})$. As with previous examples, the SAV parameter $c_0 = 10^4 E_{max} = 9590$ J/m.

Figure 22 shows the time history of displacements and velocities at points A and B of the computational model (shown in Fig. 21) obtained using the six different time discretizations described above. We observe that all six methods give accurate solutions with similar levels of error compared to the reference solutions. Figure 23 shows the von Mises stress contour at $t = 6 \times 10^{-5}$ sec, computational costs, and L^2 norms of errors. Finally, in Table 10, we list the computational gains, errors, and drifts in displacement of the different MTS methods. Once again, we observe that

implicit-explicit MTS methods presented here are able to achieve accuracy similar to explicit UTS methods with a computational gain of approximately 2.

5 Concluding remarks

A novel framework for developing new multi-time-step (MTS) methods that enable the use of different time integration schemes and different time steps is presented for solving multi-scale problems in structural dynamics and wave propagation. With this approach a problem domain is divided into several subdomains that are coupled together with Lagrange multipliers that enforce the continuity of the solution across the interface between the subdomains. However, almost all existing MTS methods enforce continuity of velocities across the interface between subdomains and consequently lead to drifts in displacements which manifest as gaps and overlaps at the interface. The proposed framework enables one to design new MTS methods to overcome this crucial limitation of existing MTS methods by simultaneously enforcing multiple continuity constraints across the interface while ensuring unconditional stability.

Almost all existing time integration methods (including all RKN and GS4 schemes) are incorporated in the proposed framework by expressing them in a new Generalized Runge-Kutta-Nyström (GRKN) formulation. With GRKN as the underlying time integration, each subdomain may be integrated using its own time integration scheme and time step. Unconditional stability of the resulting MTS method is ensured by introducing a scalar auxiliary variable (SAV) that is based on physical energy of the system. The solution process makes necessary corrections to the computed solution at every time step to keep this energy bounded as the simulation progresses and the SAV evolves in time thereby achieving unconditional stability.

New MTS methods that simultaneously impose velocity continuity and displacement continuity (VCDC) along with piecewise-linear and quadratic interpolations of Lagrange multipliers are presented as use-cases of the proposed framework. To demonstrate the performance of these new MTS methods, we present four numerical examples, the split single-degree-of-freedom (split-SDOF) problem, a 2D scalar wave propagation, the Lamb's problem, and the problem of an impact on a sandwich plate. Through these examples, we show that MTS methods lead to significant reduction in computational costs while maintaining accuracy of the solution. Furthermore, while existing MTS methods result in large drifts in displacement at the interface, the proposed VCDC MTS methods ensure zero drift in displacement and lead to more accurate solutions at a low computational cost.

Even though the computational cost of the proposed MTS methods, for comparable levels of error, is much lower than

that of the underlying time integration with a uniform time-step (UTS) approach, the convergence rate of the solution was found to be sub-optimal for some cases. Nevertheless, with the proposed framework, it is possible to design other new MTS methods that explore different coupling constraints and potentially preserve the optimal rate of convergence of the solution. A detailed study in this direction may be a fruitful avenue for future research.

Appendix A. Butcher tableau for several LMS and composite schemes expressed in GRKN form

The Butcher tableaux for single-stage GRKN schemes (i.e. $p = 1$) are shown in Tables 11, 12, 13, 14 and 15.

Table 11 The Butcher tableau for the Newmark scheme [5]

1	0	1	$1 - \gamma$	γ	1	$0.5 - \beta$	β
			$1 - \gamma$	γ	1	$0.5 - \beta$	β

Table 12 The Butcher tableau for the HHT- α scheme [8]

$1 + \alpha$	0	1	$(1 - \gamma)(1 + \alpha)$	$\gamma(1 + \alpha)$	$1 + \alpha$	$(0.5 - \beta)(1 + \alpha)$	$\beta(1 + \alpha)$
			$1 - \gamma$	γ	1	$0.5 - \beta$	β

Table 13 The Butcher tableau for the WBZ- α scheme [9]

1	α	$1 - \alpha$	$1 - \gamma$	γ	1	$0.5 - \beta$	β
			$1 - \gamma$	γ	1	$0.5 - \beta$	β

Table 14 The Butcher tableau for the CH- α scheme [10]

$1 - \alpha_f$	α_m	$1 - \alpha_m$	$(1 - \gamma)(1 - \alpha_f)$	$\gamma(1 - \alpha_f)$	$1 - \alpha_f$	$(0.5 - \beta)(1 - \alpha_f)$	$\beta(1 - \alpha_f)$
			$1 - \gamma$	γ	1	$0.5 - \beta$	β

Table 15 The Butcher tableau for the Krenk scheme [69]

0.5	0.5	0.5	$\kappa/4$	$\kappa/4$	$\kappa/2$	$\kappa^2/8$	$\kappa^2/8$
			0.5	0.5	1	$\kappa/4$	$\kappa/4$

The Butcher tableau for a two-stage GRKN scheme (i.e. $p = 2$) is shown in Table 16.

Table 16 The Butcher tableau for the Bathe scheme [16, 17, 19–21]. Parameters s_0 , s_1 , and s_2 are chosen as shown in Table 17

γ	0	1		$\gamma/2$	$\gamma/2$		γ		$\gamma^2/4$	$\gamma^2/4$
1	0	0	1	s_0	s_1	s_2	1	$s_1\gamma/2 + s_0s_2$	$s_1(\gamma/2 + s_2)$	s_2^2
				s_0	s_1	s_2	1	$s_1\gamma/2 + s_0s_2$	$s_1(\gamma/2 + s_2)$	s_2^2

Table 17 Algorithmic parameters for the Bathe schemes

Type	s_0	s_1	s_2
γ -Bathe [16, 17]	$\frac{1}{2(2-\gamma)}$	$\frac{1}{2(2-\gamma)}$	$\frac{1-\gamma}{2-\gamma}$
ρ_∞ -Bathe [20]	$(\gamma - 1)s_1 + \frac{1}{2}$	$\frac{\rho_\infty + 1}{2\gamma(\rho_\infty - 1) + 4}$	$-\gamma s_1 + \frac{1}{2}$
β_1/β_2 -Bathe [19, 21]	$\gamma(1 - \beta_1)$	$\gamma(\beta_1 + \beta_2 - 1) + 1 - \beta_2$	$(1 - \gamma)\beta_2$

The Butcher tableaux for three-stage GRKN scheme (i.e. $p = 3$) are shown in Tables 18 and 19.

Table 18 The Butcher tableau for the ρ_∞ -TSSBN scheme [30]

γ_1	0	1		0	γ_1		γ_1^2
γ_2	0	0	1	0	$\gamma_2(1-\alpha)$	$\gamma_2\alpha$	$\gamma_2(1-\alpha)(\gamma_1+\gamma_2\alpha)$
1	0	0	0	1	0	β	$(1-\beta)\gamma_2\alpha$
				0	β	$1-\beta$	$(1-\beta)\gamma_2\alpha$

Table 19 The Butcher tableau for the Noh-Bathe scheme [18]

q	0	1			q		q	$q^2/2$																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
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The Butcher tableau For a four-stage GRKN schemes (i.e. $p = 4$) is shown in Table 20.

Table 20 The Butcher tableau for the Kim's 4th-order scheme [24]

1/3	0	1				1/3				1/3	1/18					
1/2	0	0	1			1/8	3/8			1/2	1/20	3/40				
1	0	0	0	1		1/2	-3/2	2		1	1/20	9/20				
1	0	0	0	0	1	1/6	0	2/3	1/6	0	1	1/6	0	1/3	0	0
						1/6	0	2/3	1/6	0	1	1/6	0	1/3	0	0

Appendix B. Correction for Lagrange multipliers

For computing the Lagrange multipliers $\lambda(t_{\ell+\overline{\gamma}_i}^{(z)})$, one simply uses interpolation based only on λ_0 and λ_β as shown in Eq. (20). However, for specific MTS methods (e.g., PH method [46] and VC-Bathe method [70]), one employs a general interpolation for Lagrange multipliers with a correction term $\overline{\lambda}(t)$ as:

$$\lambda(t) = w_0^\lambda(t)\lambda_0 + \sum_{\beta \in \mathcal{B}} w_\beta^\lambda(t)\lambda_\beta + \overline{\lambda}(t) \quad (67)$$

where $\overline{\lambda}(t)$ accounts for the imbalance in any external forces acting on the interface between subdomains [46, 70]. For instance, the PH method uses the correction term as:

$$\overline{\lambda}(t) = C^{(1)} \left(\mathbf{f}^{(1)}(t) - w_0^f(t)\mathbf{f}^{(1)}(t_0) - w_1^f(t)\mathbf{f}^{(1)}(t_1) \right) \quad (68)$$

where weight functions $w_0^f(t)$ and $w_1^f(t)$ are obtained from linear interpolation. Likewise, for the VC-Bathe method, the following correction for Lagrange multipliers is employed:

$$\bar{\lambda}(t) = C^{(1)} \left(f^{(1)}(t) - w_0^f(t) f^{(1)}(t_0) - w_{\bar{\gamma}(1)}^f(t) f^{(1)}(t_{\bar{\gamma}(1)}) - w_1^f(t) f^{(1)}(t_1) \right) \quad (69)$$

where $w_0^f(t)$, $w_{\bar{\gamma}(1)}^f(t)$, and $w_1^f(t)$ are obtained from piecewise-linear interpolation.

Therefore, from Eq. (67), λ_0 in Eq. (28) becomes

$$\lambda_0 = \begin{bmatrix} w_0^\lambda(t_{\ell+\bar{\gamma}_1}^{(z)}) \lambda_0 + \bar{\lambda}(t_{\ell+\bar{\gamma}_1}^{(z)}) \\ w_0^\lambda(t_{\ell+\bar{\gamma}_2}^{(z)}) \lambda_0 + \bar{\lambda}(t_{\ell+\bar{\gamma}_2}^{(z)}) \\ \vdots \\ w_0^\lambda(t_{\ell+\bar{\gamma}_p}^{(z)}) \lambda_0 + \bar{\lambda}(t_{\ell+\bar{\gamma}_p}^{(z)}) \end{bmatrix}_{p^{(z)} \times 1} \quad (70)$$

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Author contributions SK worked on Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. AP worked on Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflicts of Interest The authors declare no competing interests.

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